DELAGUA CLEAN COOKING GROUPED PROJECT IN RWANDA

Document Prepared by DelAgua

Project Title	DelAgua Clean Cooking Grouped Project in Rwanda
Version	6.0
Date of Issue	03-December-2023
Prepared By	DelAgua
Contact	DelAgua Health Rwanda (Voluntary) Limited c/o Cafferata & Co., Poinciana & West Mall Drove, Freeport, Grand Bahama, P.O. Box F-42614, Commonwealth of the Bahamas E: James.Beaumont@delagua.org W: www.delagua.com

CONTENTS

1 PROJECT DETAILS

1.1 Summary Description of the Project

Summary description of the technologies/measures to be implemented by the project:

'DelAgua clean cooking grouped project in Rwanda' is a voluntary Grouped Project (GP) aimed at distributing fuel efficient improved cookstoves (ICS) to households in Rwanda. The Project Participant (PP) of the GP for all the communication, management, and ownership rights of the Greenhouse Gas (GHG) Emission Reduction Units (ERU) is DelAgua Health Rwanda (Voluntary) Limited (DelAgua) and the ICS under this GP will be distributed by DelAgua. The distribution of ICS is not mandated by Rwandan law and the Grouped Project is a voluntary initiative being run by the PP DelAgua. The program is not being implemented as a result of any government legislation or intervention. Throughout this Project Description, this grouped project may be interchangeably referred as "program". The PP plan to distribute 400,000 improved cookstove under the grouped project during first renewable crediting period. It is intended that under this GP two types of single pot portable stoves will be distributed, which are SSM S32 X stove and Burn Kuniokoa G2. Each household will receive 1 ICS.

The location of the project:

Under the GP, DelAgua will distribute the project technologies to individual households and communities within the host country Rwanda. The PAIs will occur in geographically defined regions within Rwanda. The details of the GP and PAI locations are provided in Section 1.12 of this document. All households will be in rural areas of the identified regions in section 1.12 of this document.

Implementation Dates:

Full details of implementation dates and numbers for each PAI under the GP are laid out in section 4.1 of this document.An explanation of how the project is expected to generate GHG emission reductions or removals:

The GP will follow the VCS Methodology from Sectoral Scope 3 - VMR0006 "Methodology for Installation of High Efficiency Firewood Cookstoves", version 1.1 for ER calculations. The implementation (distribution of ICS units) and parameters set for ER calculations within the GP will be monitored on a periodic basis, based on the requirements provided within the applied methodology.

A brief description of the scenario existing prior to the implementation of the project:

The scenario existing prior to the implementation of the project activity would be usage of 3-stone fire and traditional stoves, which have a very low thermal efficiency of 10%[1], and therefore consume a higher amount of firewood as fuel for cooking.

An estimate of annual average and total GHG emission reductions and removals:

The average annual and total GHG emission reduction expected from the grouped project is expected to be 1,819,332 and 14,554,657 tCO_2e, respectively, over the first 7-year crediting period.

The first monitoring period for the project activity is 11-October-2021 to 28-February-2023. The actual emission reduction achieved during current monitoring period is 1,367,923 tCO2e.

The project is neither transferred nor registered in any GHG mechanism, this project is green field project and explicitly applying the VCS registration.

Audit Type	**Period**	**Program**	**VVB Name**	**Number of years**
Joint Validation/ Verification	11-October-2021 to 28-February-2023	**VCS**	**Earthood Services Pvt Ltd**	1.386 years (506 days)
Total	**1**	**1**	**1**	1.386 years

1.2 Sectoral Scope and Project Type

Sectoral Scope 03 – Energy demand

Type II – Energy efficiency improvement projects

Activity category – non AFOLU

The project activity is designed as a Grouped Project Activity (GP) and includes multiple Project Activity Instances (PAIs) of the same activity type into one Grouped Activity. New instances can be introduced to the GP at any monitoring period, with a PAI defined as "Distribution of efficient cookstoves to a local community within a specific geographic and climatic region in the project boundary".

1.3 Project Eligibility

[1] VMR0006, Version-1.1

The Grouped Project Activity falls within the scope of the VCS Program. This is elaborated under the tables below.

Table 1: VCS Program scope

Scope	Applicability
The seven Kyoto Protocol greenhouse gases.	The GP reduces GHG emissions associated with the combustion of cooking fuels. Gases included are CO_2, CH_4, N_2O.
Ozone-depleting substances (ODS).	The GP does not involve any ODS.
Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	The ICS to be distributed in the GP applies the methodology "VMR0006 "*Methodology for Installation of High Efficiency Firewood Cookstoves"*, version 1.1." which is a VCS approved methodology.
Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval.	Not applicable.

Table 2: Excluded project activities under the VCS Program

Excluded Activity	Applicability
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional	N/A. The GP activities are energy efficiency measures and do not generate electricity from hydro-power plants/units.

grid. See the VCS Program document VCS Program Definitions for the full definition.	
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program document VCS Program Definitions for the full definition.	N/A. The GP activities are energy efficiency measures and do not generate electricity from wind, geothermal, or solar power plants/units.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).	N/A. The GP activities are energy efficiency measures and do not utilize recovered waste heat for any purposes.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass. The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO2 capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves).	As the GP activities are distribution of improved cook stoves in Rwanda to improve thermal efficiency of cook stoves used for cooking purpose. The exclusion does not apply to thermal efficiency projects i.e. cook stoves, further the project activity is located in LDC.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste	N/A. The GP activities are energy efficiency measures and do not generate electricity and/or thermal energy using fossil fuels, neither involve fossil fuel switch. The GP

containing previously used petroleum products (e.g., used plastics, oils, lubricants).	includes energy efficiency improvements in thermal applications, i.e. cookstoves.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO2e per year.	N/A. The GP activities does not include the replacement of electric lighting.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers. Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO2e per year.	N/A. The GP activities do not include the installation and/or replacement of electricity transmission lines and/or energy efficient transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	N/A. The GP reduces GHG emissions associated with the combustion of cooking fuels.

1.4 Project Design

The project has been designed to include distribution of ICS units to individual households, under multiple project activity instances within a grouped project.

Eligibility Criteria

Being a Grouped Project Activity, this project will be implemented while fulfilling the eligibility criteria provided in the Table below.

Eligibility criteria as per section 3.6.10 to 3.6.15 of the VCS Standard v4.4

Criterion	How the new project activity instances comply	Evidence
Grouped projects shall specify one or more clearly defined	Each PAI will have clearly defined geographic areas	PAI 1 to PAI 4 has been implemented in Rwanda.

geographic areas within which project activity instances may be developed. Such geographic areas shall be specified using geodetic polygons	and such area will be specified using geodetic polygons.	PP has submitted geodetic polygons to VVB and added details in section 1.12 of the PD MR.
Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances. The initial project activity instances are those that are included in the project description at validation and shall include all project activity instances currently implemented on the issue date of the project description.	Baseline scenario and demonstration of additionality shall be based upon the initial project activity instances	Baseline scenario and additionality has been demonstrated for PAI1 to PAI4 which is included at the time of validation in section 3.4 and 3.5 of the PD MR.
Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area.	The grouped project includes only one project activity.	The PAI1 to PAI4 includes distribution of improved cookstoves to households that are dependent on traditional cookstoves using woody biomass for cooking.
The baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project.	Baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project	Baseline scenario has been determined for PAI1 to PAI4 for Rwanda. Details have been added in section 3.4
The additionality of the initial project activity instances shall be demonstrated for each designated geographic area, in accordance with the methodology applied to the project	Additionality for a project activity instance shall be determined for each designated geographic area, in accordance with the methodology applied to the project	Additionality has been determined as per the designated methodology. Details have been added in section 3.5.

Where factors relevant to the determination of the baseline scenario or demonstration of additionality require assessment across a given area, the area shall be, at a minimum, the grouped project geographic area. Examples of such factors include, inter alia, common practice; laws, statutes, regulatory frameworks, or policies relevant to demonstration of regulatory surplus; determination of regional grid emission factors; and historical deforestation and degradation rates	All the factors relevant to the determination of baseline or demonstration of additionality shall be done for the geographical boundary of the grouped project	Assessment of baseline and demonstration of additionality for PAI-1 to PAI-4 has been done for Rwanda.

Eligibility criteria as per section 3.6.16 and 3.6.17 of the VCS Standard v4.4:

S No	Criterion	How the new project activity instances comply	Evidence
1	Project Activity Instances (PAIs) must meet the applicability conditions set out in the applied methodology.	*Please refer the subsections below.*	*Please refer the subsections below.*
1a	Project activities shall be implemented in domestic premises or in community-based kitchens.	Each PAI will distribute technology that is appropriate for domestic use to household end-users	a) Manufacturer's specifications. b) Project ICS distribution database containing unique household end user information. The ICS distributed under PAI-1, PAI-2, PAI-3 and PAI-4 are for domestic use i.e. for cooking in individual

			households. The evidence for the same is submitted to VVB.
1b	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	ICS distributed in each PAI will have a thermal efficiency of at least 25%, and will be confirmed by the end user that they were using the firewood as the cooking fuel in the project's baseline. Each PAI will record the baseline and project fuel use to ensure that wood fuel is the sole source of energy.	1.Manufacturer's specifications. 2. Information recorded within the baseline KPT and project monitoring surveys on the fuel usage. The PAIs included under grouped project activity has distributed Kuniokoa generation 2 stoves having thermal efficiency of 51%. The supportive for the same is submitted to VVB.
1c	Both 'Projects' and 'Large Projects' can use the methodology	PAIs will comply with the GP requirements	The GP is defined as a "Large Project" according to the guidelines set out in the VMR0006 methodology and can therefore apply the methodology. No further assessment required at PAI level.
1d	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	This has been demonstrated at grouped project level that demonstrate that Non-renewable biomass has been used in the project region since 31 December 1989.	No further assessment required during inclusion of PAI.

1e	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Biomass residues are not processed as a fuel to be used in the GP. The project activity does not involve fuel switch from baseline case. End users will be asked to state their source of energy and baseline appliance at the point of distribution and during the project monitoring. Above information will be stored in the monitoring database.	1. Monitoring surveys. 2. Project ICS distribution poster specific to the end-users (sample photographs) . The ICS distributed under PAI-1, PAI-2, PAI-3 and PAI-4 replaces wood fuel based traditional stoves, the same has been confirmed and recorded during ICS distribution and also in monitoring survey. The details submitted to VVB.
1f	The Project Design Document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo)	To avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient Each PAI will have system for avoiding double counting with data being collected via a smartphone	1. Project (GP) Distribution database can be checked to verify the distribution approach, along with the distribution system (includes both plan and execution steps) established by the PP. The distribution database is an electronic document system. 2. The VCS GP ICS units have unique engravings or text specifically mentioning the unit is

		app (where possible), or via paper forms, including: 1. Date of distribution 2. A unique serial number[2] on each technology 3. Available end-user data: name, phone number, address/physical location, ID number	strictly "not for sale" along with the DelAgua logo and will have a unique identification number, which ensures that same ICS is not accounted in other PAI. For PAI-1, PAI-2, PAI-3 and PAI-4 the distribution record and written attestation from the project proponent confirming that the ICSs distributed are not part of any other CDM or Voluntary GHG programme is provided to VVB.
1g	The Project Design Document shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The End User Agreement will confirm that the ownership rights of the ER/VCUs generated by the PAI reside with the PP. The signature of each end user confirming their understanding of this agreement will be collected at the point of distribution. The End User Agreement will be in English and translated into a local language where required. If necessary, it will be read out to the End User.	a) End users will be asked to sign the End User Agreement and this information will be a part of the agreement, and then stored in the monitoring database. b) Each PAI will include a statement on the uniqueness of the Project and by listing other

[2] There are no similar programs running in the host country Rwanda where the same ICS models are distributed by another entity under different program(s). If this is found during the project ICS monitoring or verification, it shall be ensured that double counting is avoided by using distinct nomenclature for the unique ID code of the ICS compared with the different programs, and also making sure that it has a distinct color, label, or marking.

		All rights to distribute the delivered stoves and claim the resultant GHG emissions reductions belong to the PP. The GP will not include any PAI that has been rejected by another GP, or PA-DDs of any other GHG programs.	similar activities (from other GHG programme registries) that are taking place within the project boundary to demonstrate its uniqueness. The beneficiaries under PAI-1, PAI-2, PAI-3 and PAI-4 have confirmed their agreement through signature on posters during receiving of ICS. The distribution database shared with VVB.
2	Use the technologies or measures specified in the project description.	Only ICS that conform with the GP description are to be distributed in the project. The ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance.	Manufacturer's specifications. The ICS distribution under PAI-1, PAI-2, PAI-3 and PAI-4 are Kuniokoa generation 2 stoves, which is as per technology description of the grouped project.
3	Apply the technologies or measures in the same manner as specified in the project description.	Only ICS that conform with the GP description are to be distributed in the project. The ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance.	Manufacturer's specifications. The ICS distributed under PAI-1, PAI-2, PAI-3, PAI-4 reduces fuel wood consumption due to better thermal efficiency and replaces the baseline traditional stoves, the same can be checked with distribution record and sample survey results.

4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All new PAIs will be implemented only in regions within the geographic borders of Rwanda subject to the same baseline scenario determined in Section 3.4 in this Project Description. The geographical location of the project ICS can be traced through the information provided in the distribution database, which includes the location data of the ICS distributed.	ICS Distribution database For PAI-1, PAI-2, PAI-3 and PAI-4 records distribution (end user registration database) with clear delineation of project boundary i.e. host country of Rwanda is provided to the VVB for their reference.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All new PAIs will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no government mandated programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances will comply with the positive list as it satisfies criterion 1 where it meets all the applicability conditions of the methodology. Where the project activity installs or distributes stoves at zero cost to the	Step1: Regulatory Surplus There is no mandatory legal requirement for installation of cookstoves in households/ institutions of Rwanda. Step 2: Positive List Further, the PP distributing ICS free of cost to end users, hence there is no revenue stream than sale of VCUs, hence the project activity instances are automatically additional. The above can be checked with end user agreement and declaration from PP on the same. The PP has provided end user agreement and

		end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.	declaration to VVB to support free distribution of ICS.
6	Where a capacity limit applies to a project activity included in the project, no PAI shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: Each device that exceeds one percent of the capacity limit shall be identified. Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters. None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.	The GP does not have any specified capacity limits applicable, since there are no capacity limits defined within the applied methodology, VMR0006, ver. 1.1. Therefore, this eligibility criteria is not applicable.	N/A
Eligibility conditions specific to inclusion of New PAIs			
7	The new PAI must occur within one of the designated	The distribution of the ICS will be based on a MoU	1. Sample list of the beneficiaries

	geographic areas specified in the project description	signed with the Government of Rwanda, and therefore, the project activity only distributes to the beneficiaries that are provided in Ubudehe based list of the districts. There is no way this program/GP can distribute ICS in the boundaries of the neighbouring countries of Rwanda. This requires passing of the project ICS through customs and other procedural regulations.	2. The MoU with the Rwanda government The evidence in line with above share with VVB for PAI-1, PAI-2, PAI-3 and PAI-4.
8	The new PAI must comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	The eligibility criteria set must be complied by the PAIs at all points in time throughout the crediting period. If at any point, it is found that the PAI is not complying with the set of eligibility criteria, then it may be excluded from the crediting for the duration of the monitoring period, until it is able to demonstrate the compliance of the eligibility criteria.	The MR will act as the documentary compliance summary. However, for each specific eligibility criteria (as laid down above), the supporting evidence will be provided by the PP. All the PAIs i.e. PAI-1, PAI-2, PAI-3 and PAI-4 meets the eligibility criteria for inclusion under grouped project activity.
9	The new PAI must be included in the monitoring report with sufficient technical, financial, geographic and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	The new PAI will include all the technical, financial, geographic and other relevant information related to the compliance of the applicable set of eligibility criteria & also enable sampling by the VVB.	The MR for the specific monitoring period will act as the supporting evidence for the stated condition/requirement. The PAI-1, PAI-2, PAI-3, PAI-4 are part of VCS grouped project validation/verification, hence no further assessment.

10	The new PAI must be validated at the time of verification against the applicable set of eligibility criteria.	The new PAI(s) shall be part of the monitoring, and therefore, will be validated against the criteria specified, by a VCS accredited/approved VVB.	The Verification Report based on the VCS template, shall be issued by the approved VVB. This will act as the supporting evidence against this requirement. The PAI-1, PAI-2, PAI-3, PAI-4 are part of VCS grouped project validation/verification, hence no further assessment.
11	The new PAI must have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	The PAI related evidence on the project ownership will be in the form of a distribution end user receipt poster. This will be signed by the end user, and a photograph of him/her holding the poster along with the ICS will be saved in the PAI database.	1. A photograph of the PAI distribution first end user holding the poster along with the ICS. 2. A scan of the poster, showing the first distribution date. 3. The distribution database of the GP stating the start date. Waiver of carbon credits from the end users are submitted to the VVB for PAI-1, PAI-2, PAI-3, PAI-4
12	The new PAI must have a start date that is the same as or later than the grouped project start date.	The new PAI will have a start date later than the GP start date.	This can be verified through the GP distribution database, wherein each new PAI can be verified for its start date. The start date of grouped project activity is same as start date of PAI. The distribution record shared

			with VVB to support start date of PAI.
13	The new PAI must be eligible for crediting from the start date of the instance through to the end of the project crediting period (only).	The PAI will be only eligible for the crediting from its start date through to the end of the project crediting period (only).	1. The start date of the PAI can be checked through the distribution database. 2. The end date of the GP crediting shall be referenced from the VCS Project Registry webpage. 3. The overlap of monitoring periods may be checked through the previous monitoring reports (available publicly on VCS project registry webpage for those periods). The PAI-1, PAI-2, PAI-3, PAI-4 are eligible for crediting from the start date of the instance through to the end of the project crediting period
14	The new PAI shall not be or have been enrolled in another VCS project .	It will be ensured that the PAI that is being enrolled under the VCS GP was not previously enrolled under any other VCS project, and will not be enrolled under other VCS project if it is removed from the VCS GP.	Since, for PAI-1, PAI-2, PAI-3 and PAI-4 project proponent is the implementer, no further approval is required, however for the cases, where implementer other than PP, an approval on the same shall be given by PP in order to include the instance into this grouped project.
15	The new PAI shall adhere to clustering requirement that states the project proponent	It will be ensured that PAI included within 10 kilometers of another	PAI-1, PAI-2, PAI-3 and PAI-4 involves distribution of improved cookstoves by

	shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent	instance of the same project activity by the same PP is in a singular project	same PP within 10 kilometers hence included in the same project. There are no other project instances by the same PP distributed within 10 kilometers for the same project activity.

1.5 Project Proponent

Organization name	DelAgua Health Rwanda (Voluntary) Limited (DHRV)
Contact person	James Beaumont
Title	Mr.
Address	c/o Cafferata & Co., Poinciana & West Mall Drove, Freeport, Grand Bahama, P.O. Box F-42614, Commonwealth of the Bahamas
Telephone	+250-790002992
Email	James.Beaumont@delagua.org

Organization name	DelAgua Project Pte. Ltd.
Contact person	Neil Mcdougall
Title	Mr.
Address	c/o c/o 38 Beach Road, South Beach Tower, #29-11, Singapore 189767
Telephone	+352 661 939 304
Email	Vcsprojects@delagua.org

1.6 Other Entities Involved in the Project

N/A. There are no other entities involved in this Grouped Project activity.

1.7 Ownership

DelAgua Health Rwanda (Voluntary) Limited is the owner of the VCS grouped project activity.

During the distribution of the improved cookstoves, the participating households sign an end user agreement between customer and Delagua. The end user agreement has customer information, unique identification number, product details etc. The agreement follow the requirements of para 3.7.1 of the VCS Standard ver4.4 which states the evidence to establish project ownership should be "An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals which vests project ownership in the project proponent."

The agreement confirms that the ownership rights of the VERRA project and the carbon assets generated from this project lie with the project proponent. The customer under the End User Agreement "releases all rights to the greenhouse gas reductions and carbon credits produced by the use of the clean energy product in the favour of Delagua and agree to not sell or transfer the GHG or carbon credits to any other third party or use these credits for any other purposes

This will be evidenced through the below:

- DelAgua has executed a Memorandum of Understanding with the Government of Rwanda, whereby DelAgua is assigned responsibility for managing and implementing the project and is recognized as the sole direct recipient of all carbon credits derived from it.
- Manufacturer agreements confirm that DelAgua is the sole owner of all technologies purchased under this project. These agreements further confirm the manufacturer's understanding that the carbon credits associated with use of the technologies shall be within the jurisdictions of DelAgua.

During the distribution of the ICS, the participating household will sign an End User Agreement to confirm that the ownership rights of the carbon assets generated from this project lie with the project proponent.

1.8 Project Start Date

The start date for this Grouped Project is 11-October-2021, which is the date of commissioning of first batch of ICS under grouped project.

1.9 Project Crediting Period

The project crediting period will be seven years, twice renewable, for a maximum total of 21 years.

The first crediting period will be 11-October-2021 to 10-October-2028, including both the dates.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	☒

Year	Estimated GHG emission reductions or removals (tCO_2e)
11 Oct 2021- 31 Dec 2021	379,873
01 Jan 2022-31 Dec-2022	2,243,669
01 Jan 2023-31 Dec-2023	2,160,906
01 Jan 2024-31 Dec-2024	2,082,325
01 Jan 2025-31 Dec-2025	2,044,721
01 Jan 2026-31 Dec-2026	2,033,761
01 Jan 2027-31 Dec-2027	2,001,708
01 Jan 2028-10 Oct-2028	1,607,694
Total estimated ERs	**14,554,657**
Total number of crediting years	7
Average annual ERs	**1,819,332**

1.11 Description of the Project Activity

Distribution plan:

The distribution of ICS will occur at centralized distribution points, where households from surrounding villages will be educated on the proper use and maintenance of the technologies. Shortly following the initial distribution, CHWs will visit each household that received a technology to deliver the training again, ensuring that all members of the household are aware of proper use and maintenance.

The ICS distribution is planned in close coordination with the local government offices, e.g. District Offices, Sector Offices, and so forth. There is a structured plan through which the distribution activity is scheduled and then implemented on ground.

Project activity description (including the technologies or measures employed) and how it/they will achieve net GHG emission reductions or removals.

The project activity included in this project description is the distribution and installation of high efficiency ICS systems in PAIs within Rwanda.

The ICS units will be based on a "rocket stove" design which reduce GHG emissions from biomass burning through improved combustion efficiency of wood fuel and decreased wood fuel consumption. The project stoves improve heat transfer to cooking pots through the insulated walls and design of the stove top when compared to traditional cooking methods over open- and three-stone fires. The project stoves directly decrease wood fuel demand and reduce the pressure from wood cutting within the project boundary. A 30-50% saving of firewood when compared to three-stone fire cooking has been reported in certain models. The time spent foraging for wood fuel is thereby also reduced, saving time for especially women to engage in other economic driven activities.

ICS technologies planned under the GP for distribution:

The following ICS models are envisaged to be distributed in the initial PAIs –

Model-1[3]:

Name of the stove – Kuniokoa Generation 2

Stove manufacturer – Burn

Stove Type – Natural Draft

Stove Body – Stainless Steel

Stove height – 32.1 cm

Stove diameter – 28.2 cm

Thermal efficiency (with pot skirt) – 51.3%

Thermal efficiency (without pot skirt) – 43.4%

High Power Output – 2.9 KW

Stove lifetime – 10 years

[3] Manufacturer specification and third party lab report submitted to VVB

Figure 1: Kuniokoa Stove Model reference picture

Model-2[4]:

Name of the stove – SSM S32-X

Stove manufacturer – SSM

Stove Type – Natural draft

Stove Body – Stainless Steel

Stove height – 27 cm

Stove diameter – 32 cm

Thermal efficiency with Pot Skirt – 50.1%

Thermal efficiency without Pot Skirt – 45.7%

High Power Output – 2.9 KW

Stove lifetime – 10 years

Figure 2: S32-X Stove Model reference picture

[4] Manufacturer specification and third party lab report submitted to VVB

List of main manufacturing/production technologies arrangements, systems and equipment involved.

Detailed application and information on the ICS technology that is going to be distributed under the new PAIs will be described in the upcoming MRs, elaborating the information under relevant section of the MR for the newly included PAIs. PP will source the best available technologies for distribution under the GP.

At the time of validation and verification 398,608 BURN Kunikoa stoves had been distributed in rural areas of Rwanda.

PAI-1: 24,274

PAI-2: 250,029

PAI-3: 99,412

PAI-4: 24,893.

The ICS model selection at PAI level will be based on key attributes such as thermal efficiency of minimum 25%, durability, cultural acceptability, and manufacturing quality and warranty. From time to time, requirement gathering will be done on a continuous basis, seeking comments from the end users (through monitoring surveys, feedback requests on call center, feedback via register available at different sites, and anonymous feedback through CHWs), and based on these user feedbacks, newer models will be introduced to the GP, and new PAIs will be designed to accommodate for their variability in terms of design, specifications, and geographies within the host country (wherever required).

It will be verified by the PP that the project technologies are, at a minimum, meeting the technical specifications and requirements set in the respective methodologies (under the eligibility criteria).

At present, DelAgua is sourcing the ICS from different manufacturers, deploying them at its central warehouse through imports, and then distributing them in a scheduled manner. Each distribution involves various steps, which are elaborated in the relevant section of this document. In this whole process, quality of the equipment being distributed will be checked on spot basis. Each lot of the distribution batch will undergo spot checks, as it is an integral part of Quality Assurance at DelAgua. The ICS units that will be found as faulty, will not be distributed to the end users. At present, only rocket stove models are planned for the upcoming distribution PAIs, as their distribution is found both beneficial to the locals (usage and maintenance) as well as a successful one from the implementation point.

The types and levels of service

Level of Service: PAIs will provide ICS based on the 'rocket stove' design, delivering a level of service at least equivalent to the baseline appliance (open fire used for household cooking).

Type of Service: The ICS technology will deliver a specified high-power thermal efficiency of at least 25% and evidenced by a manufacturer's specification and WBT certificate. This is a significant improvement on the baseline open fire thermal efficiency default of 10%.

List of facilities, systems and equipment in operation under the existing scenario prior to the implementation of the project.

Under the baseline scenario, households use non-renewable woody biomass for their fuel needs for cooking. The baseline appliance being replaced by the project is an open fire (typically based on the three-stone concept).

1.12 Project Location

For each PAI, the project boundary will be defined as being within the geographic borders of Rwanda. The project boundary for each Project Activity Instance is the physical, geographical location of the ICS distributed under it within the geographical boundaries of Rwanda.

Also, each individual distribution under the project technology will record the geographical location related data (e.g. latitude/longitude data), which can be verified from the recorded data (monitoring surveys).

KML files will be provided as supporting documents to demonstrate the extent and the geographical coverage of the PAIs within the GP. All the KML files will indicate the regions within the geographical boundaries of the host country Rwanda.

The project boundary is defined by the following GPS coordinates:

2° 50' 23.8992" S and 28° 51' 42.1452" E

1° 2' 50.82" N and 30° 53' 56.6592" E

Fig. Project Location

1.13 Conditions Prior to Project Initiation

The baseline scenario is the same as the conditions existing prior to the Grouped Project initiation.

Majority of people in the host country still use traditional cookstoves or open fires which consume a large amount of fuelwood. The project aims to reduce the environmental and social burdens caused by inefficient biomass cooking by introducing energy efficient cookstoves to the individual households. The current scenario as established is continuation to use traditional wood fuel-based stoves to meet thermal energy demand for cooking.

For further information regarding this, please refer Section 3.4 (Baseline Scenario) of this document.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

A review of the national energy policies of the Rwanda shows that there are no mandatory laws, policies or requirements mandating the use of ICS. Though the Government of Rwanda's (GoR) Energy Sector Strategic Plan (2013-2018) identifies dissemination of improved cookstoves (ICS) as one way to reduce fuel wood consumption. The government promotes the use of improved ICS, however it's not mandatory.

The GP complies with applicable local laws and regulatory frameworks. Under Rwandan law and regulation, none of the activities or processes relating to the manufacture or distribution of Improved cookstoves require Environmental Impact Assessments not meet any of the screening criteria in Appendix 2, page 40 of the EIA guidelines[5].

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

No. This proposed VCS Grouped Project is neither registered nor is seeking registration under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

Not applicable, as this project is a new Grouped Project activity and has not been rejected by other GHG programs.

1.16 Other Forms of Credit

[5] General Guidelines and Procedure for Environmental Impact Assessment, Republic of Rwanda, November 2006 (See filename: RW_EIA_Guidelines_Final_versionl_Nov_2006)

1.16.1 Emissions Trading Programs and Other Binding Limits

The grouped project does not reduce GHG emissions from activities that are included in any emissions trading program.

1.16.2 Other Forms of Environmental Credit

The VCS GP has not sought or received another form of GHG-related environmental credit, including renewable energy certificates.

1.16.3 Supply Chain (Scope 3) Emissions

Not applicable

As per VCS Standard version 4.4, this section (Scope 3 emissions inventories) is effective 1st January 2024 onwards hence this section is not applicable for the current monitoring period i.e., 11-October-2021 to 28-February-2023..

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The implementation of PAIs – 1, 2, 3, & 4 led to the following Sustainable Development contributions:

The GP is contributing towards sustainability, which is also verifiable through the SDG indicators selected for the specific activities that are going to be monitored for their fulfilment.

Following sustainable contributions were part of the PAIs – 1, 2, 3, & 4 (Distribution of ICS in Rwanda):

1. SDG-1: No Poverty

   Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG. In this case, the yardstick for access to basic services was kept at RWF 100,000/HH(household)/month. Based on DelAgua's experience in Rwanda, this is the value below which access to basic cooking services is not possible.

   Relevant SDG Indicator: 1.4.1 Proportion of population living in households with access to basic services

   Available Data: The parameter that was monitored to calculate the number of HHs living without access to basic services was the number of beneficiary HHs. This is because the number of HHs that are unable to purchase any improved cooking devices can be considered as the ones living without access to the basic services such as clean cooking. The list of HHs which are to be distributed with the project ICS is provided by the government of Rwanda, and is based on the economic classification list of the population of Rwanda.

Monitoring Requirement(s): The new project ICS beneficiary/user will be considered as living without the access to clean cooking, and the final value of number of such people using the project ICS will be considered towards the estimation of this SDG impact. Clean cooking is considered as a basic service, and access to it is an addition to the SDG impact.

2. SDG-3: Good Health and Well Being:

Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG, specifically the SDG indicator. This SDG relates to health, and DelAgua's project ICS is improving health for every individual who is receiving it.

Relevant SDG Indicator: 3.9.1 Mortality rate attributed to household and ambient air pollution

Available Data: Regarding indicator 3.9.1, at every project ICS beneficiary HHs, ones who are experiencing any improvement (or no improvement or deterioration) in their ambient air quality as compared to pre-project scenario will be considered as an input.

Monitoring Requirement(s): The data was collected at the time of monitoring survey. Beneficiary HHs for VCS GP will be sampled for monitoring as per monitoring plan, and these are then requested to respond to survey questions, which is then recorded within the monitoring survey database.

3. SDG-5: Gender Equality

Approach: In this case, the number of ICS beneficiary HHs will be questioned over their time spent for collecting fuel (firewood).

Relevant SDG Indicator: 5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location

Available Data: Monitoring Survey for VCS GP includes question on time spent on collection of fuel.

Monitoring Requirement(s): The value is recorded in the monitoring surveys, and then saved in the survey database.

4. SDG-7: Affordable and Clean Energy

Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG, specifically the SDG indicator.

Relevant SDG Indicator: 7.1.2 Proportion of population with primary reliance on clean fuels and technology

Available Data: Distribution Master Database includes the data for this, where each HH where project ICS is distributed, acts as an input. The data within the database is updated on a continuous basis.

Monitoring Requirement(s): Distribution Database is exported for the monitoring period, and then the number of beneficiary HHs are multiplied with the average HH size of Rwanda (which is 4.3 in 2021). This gives the number of people who are now benefiting from the clean and affordable energy for the Monitoring Period.

5. SDG-8: Decent Work and Economic Growth

Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG, specifically the SDG indicator.

Relevant SDG Indicator: 8.3.1 Proportion of informal employment in total employment, by sector and sex

Available Data: The salary slips for the CHWs act as the data source since these are not on DelAgua's payrolls. However, there are different data sources (number of mobile phones assigned to individuals during distribution, number of CHWs trained for a specific distribution event), which are used to total the CHWs working for DelAgua.

Monitoring Requirement(s): This data is continuously collected and is obtained from Training and Account teams.

6. SDG-12: Ensure Sustainable consumption and production pattern

Approach: Here, each household is trained for sustainable practices, best usage practices for ICS usage, and utilization of fuel resources. Apart from this, the project being one of its kind, provides education and training to 5000+ Community Health Workers (CHWs) engaged with us for distribution and end-user engagement.

Relevant SDG Indicator: 12.8 By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature

Available Data: (i) Number of project ICS beneficiary HHs who were visited by DelAgua staff for HH visit surveys are used to calculate the parameter. This data is continuously recorded in the distribution master database.

(ii) Number of CHWs that were trained for the project activity implementation (distribution and surveys).

Monitoring Requirement(s): The distribution master database and the CHW training data is used to calculate the number of people trained under the project activity.

7. SDG-13: Climate Action

Approach: For this SDG, there are no separate monitoring requirements. We declare the GHG Emission Reductions achieved by project in the monitoring period. This is verified by a VCS accredited VVB, and the resultant VCUs are then issued, acting as the net impact contributed towards this SDG.

Relevant SDG Indicator: Not applicable.

Available Data: This can be accessed from the Monitoring Reports that are publicly available.

Monitoring Requirement(s): All the monitoring requirements are already detailed in the Monitoring Reports and corresponding ER Sheets.

8. SDG-15: Life on Land

Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG, specifically the SDG indicator.

Relevant SDG Indicator: 15.4.2 Mountain Green Cover Index

Available Data: (i) Baseline (pre-project) survey data, (ii) Monitoring Surveys (periodic), (ii) Registered PD&MR and ER Sheet

Monitoring Requirement(s): This parameter is fixed for per HH level firewood consumption, which is then multiplied with monitored values (such as usage rate), other relevant values as per the equations provided in the methodology applied for VCS GP. Based on the calculations, per HH level firewood savings are calculated.

The SDG impacts of the distribution program and monitoring related discussion is provided in the below sections.

1.17.2 Sustainable Development Contributions Activity Monitoring

Summary description of project activities implemented during the monitoring period that result in SD contributions (i.e., technologies/measures implemented, activity location):

The VCS GP involved PAIs that led to a combined Sustainable Development action through the distribution of ICS units across rural households in Rwanda. The distribution of ICS leads to several direct and positive impacts into the lives of these beneficiary households.

An explanation of how project activities result in the SD contributions described in Table 1 of this report:

The VCS GP impacts the lives of the households in many ways. These are then reviewed by the PP, and it was found out that these impacts from the project activity at every household level may be documented with the help of UN SDGs. Each impact is then mapped against a relevant SDG indicator, which is then planned for the monitoring. Survey questionnaire used for monitoring the project activity includes questions on the SDG impacts, and the responses from the end user acting as respondent to the monitoring survey are then recorded against those. These survey responses recorded are saved in the online survey form, and these are then analysed for 90/10 rule as per the CDM sampling standard requirements.

Identification of which SD contributions described in Table 1 of this report contribute to achieving any nationally stated sustainable development priorities, including any provisions for monitoring and reporting the same:

The VCS GP support the nationally stated sustainable development priorities of the host country, and result in a reduction in the deforestation due to firewood collection and usage for cooking purposes. This supports Rwanda's goal[6] of reducing the percentage of households using wood fuel and charcoal for cooking from 83.3% in 2014 to 42% in 2024. This directly impacts the SDG-15, indicator 15.4.2 Mountain Green Cover Index, by reducing the firewood cutting from the forests and consequently increasing the forest cover compared with the baseline levels. However, there are no provisions from the host country government on the monitoring and reporting of any of those SDGs.

[6] https://www.undp.org/rwanda/rwanda%E2%80%99s-health-and-forests-need-clean-break-cook-stoves

Table 1: Sustainable Development Contributions[7]

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	1.4	Indicator 1.4.1: Proportion of population living in households with access to basic services	Implemented activities to increase	Cooking is considered a basic service, as food is considered a basic need of humans and other forms of life. Therefore, improved cooking access is a basic need as well. Under current monitoring period 4 PAIs included wherein total 398,608 households provided with improved cookstoves and 100% ICS operational as below PAI-1: 24,274 HHs PAI-2: 250,029 HHs PAI-3: 99,412 HHs PAI-4: 24,893 HHs So, a total of 398,608 households now have access to improved cookstoves when compared with the baseline.	398,608 households got access to ICS As this is the first monitoring period for this GP, the cumulative SDG impacts are the same as the monitored ones.

[7] The SDG contribution from the VCS GP will be provided at the time of Validation. Currently, the monitoring is planned for the project activity.

2)	3.9	3.9.1 Mortality rate attributed to household and ambient air pollution	Implemented activities to decrease	Mortality rate attributed to HH and ambient pollution will decrease for those HHs using the DelAgua improved cookstove. The improved design specifications of the project ICS model(s) ensure there is very less pollution in the households. Out of the total population of Rwanda, a total of **1,715,915** people now have less risk towards respiratory ailments due to smoke inside their homes due to the implementation of VCS 4150 (for first monitoring period).	Total **1,715,915 individual benefited** As this is the first monitoring period for this GP, the cumulative SDG impacts are the same as the monitored ones.
3)	12	12.8 By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature	Implemented activities to increase	Due to grouped project the team involved in the distribution of ICS being provided training, who further educates the end user HHs about the sustainable cooking practices and awareness on climate change impacts. i) The number of CHWs that were educated (trained for distribution, 6-monthly household visits, surveys, and training on sustainable cooking practices to be disseminated to end user households) by the Grouped Project (for PAI-1, PAI-2, PAI-3 & PAI-4) by the end of M.P.-1 are 4,685. ii) The number of end users per household that are educated for the stove usage and apply sustainable cooking practices are equal to the exact number of households where project ICS are distributed. Total number of HHs educated are 398,608. PAI-1: 24,274 HHs PAI-2: 250,029 HHs PAI-3: 99,412 HHs PAI-4: 24,893 HHs	Total 4,685 CHW workers trained Total 398,608 HHs being educated As this is the first monitoring period for this GP, the cumulative SDG impacts are the same as the monitored ones.

4)	5.4	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	Implemented activities to decrease	The % of HHs where time spent on collecting fuel and cooking meals has decreased with the implementation of this project activity. During monitoring surveys, it was identified that 96% of respondents acknowledge that they experienced a decrease in the time spent on collection of cooking fuel (firewood). This, when multiplied with the total number of end user HHs, give 398,608 (rounded down to nearest whole number). So, 383,088 HHs spending less time on unpaid domestic work (i.e. collection of firewood for cooking).	383,088 beneficiaries saving time As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same as the monitored ones.
5)	7.3	By 2030, double the global rate of improvement in energy efficiency	Implemented activities to increase	For this GP, the baseline scenario is the households using traditional cook stoves. This project increases the adoption rate of efficient cookstoves. The outcome of SDG 7 will be measured by the number of efficient cookstoves distributed during the project. During current monitoring period below are the number of households got access to improved cookstoves PAI-1: 24,274 HHs PAI-2: 250,029 HHs PAI-3: 99,412 HHs PAI-4: 24,893 HHs	Number of project households predominantly using efficient cooking devices such as Improved Cook Stoves. 398,608 HHs during the monitoring period. As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same as the monitored ones.

6)	8.3	8.3.1 Proportion of informal employment in total employment, by sector and sex	Implemented activities to increase	The project activity has a database of CHWs working for it. This database is used to provide the actual number of CHWs working for the project activity. The number of permanent employees working on the DelAgua payroll will not act as a part of this indicator since it is only focused on the informal employment. However, the number of estimated people who worked in the manufacturing of these ICS, and those working for the delivery and transportation of these ICS will be considered towards the indicator. During M.P.-1, the project activity provided informal employment to around 4,685 people.	Around 4,685 employments direct/indirect both. As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same as the monitored ones.
7)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	The emission reductions will be quantified as per the methodology applied. During current monitoring period the 4 PAIs together have reduced 1,367,923 tCO_2e.	GHG Emissions 1,367,923 CO_2e saved As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same the monitored ones.

8)	15.4	Indicator 15.4.2 Mountain Green Cover Index	Implemented activities to increase	The implemented project activity led to a decrease in the deforestation in the mountainous forest ecosystem. This eventually led to an increase in the mountain green cover index compared with the baseline. From the ER calculations made for PAI-1 to PAI-4, as per monitoring survey of project ICS is the calculated value of firewood savings. If we multiply this with the number of HHs, we'll get the amount of firewood saved by the GP. M.P.-1. PAI-1: 5,759.75 ton PAI-2: 448,633.83 ton PAI-3: 252,314.72 ton PAI-4: 33,309.90 ton	Total wood fuel saving during current monitoring period is 740,018 tons. As this is the first monitoring period for this GP, the cumulative SDG impacts will be the same the monitored ones.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

N/A as there is no other bearing on the eligibility of the project.

2 SAFEGUARDS

2.1 No Net Harm

The project undertakes the distribution of improved cook stoves. As per the Laws of Rwanda, the project undertaking such activities does not require any Environmental Impact assessment. This is as per Ministerial Order No. 004/2008: Ministerial Order establishing the list of works, activities and projects that must undertake an environment impact assessment[8]. The fact that this project activity does not need environmental assessments, supports that it does not cause environmental harm.

The project, on the contrary, has several positive social and environmental benefits due to its implementation. The project leads to improved indoor air quality, lower air emissions, and also positive health impact.

2.2 Local Stakeholder Consultation

The local stakeholder's consultation meeting for the VCS GP was planned and conducted to at grouped project activity level, wherein participation expected from local stakeholders including NGOs, Academia, other project developers, and any other entity who concerns itself with the DelAgua clean cookstove distribution.

The meeting intended to do the following:

[8] https://www.elaw.org/rwanda-environmental-impact-assessment-law-and-regulations

1. To gather local people (the expected beneficiaries) together and inform them about the VCS grouped project activity.

2. To make them aware about the Carbon Finance without which this VCS grouped project could not be established.

3. To make the local people understand the benefits of clean cooking.

4. To demonstrate the working of sample cookstove that is aimed for the distribution.

5. To collect their feedback, queries, concerns, and respond those at the meeting itself.

The relevant stakeholder for LSC meeting being invited using various mode to get the participation/feedback which includes publication in newspaper “The New Times” dated 03 & 13 February 2023, email sent to identified NGOs and other organizations dated 04 & 13 February 2023 and advertisement in radio channels. The local stakeholder consultation meeting was held on 01 March 2023 from 9.00 AM to 11.00 AM at Gorillas Golf Hotel, Nyautarama, Rwanda.

The agenda of the meeting was to showcase the VCS grouped project being implemented and invite local stakeholders other than the potential end users to understand about the project, submit their grievances, and raise their concerns, wherever required. Below are the steps followed during stakeholder consultation meeting

1. The meeting started with presentation on grouped project activity and associated benefits to society, ecology and economic aspect due to use of improved cookstoves.

2. The sample cookstove unit, which was being distributed under VCS 4150 implementation (Burn Kuniokoa), was demonstrated to the local stakeholders present for its usage.

3. During the meeting, stakeholders were invited to raise their queries and concerns. These were duly responded to during the meeting itself.

Below are the queries raised by the attendees

Q.1. Whether current grouped project activity of DelAgua involves only improved cookstove distribution?

Ans: The project activity involves only Improved cookstove distribution.

Q.2. Which districts/region the project is planned under the current project activity?

Ans: The PP representative explained that the project boundary for the grouped project activity is host country Rwanda and various districts will be targeted where ICS penetration is still less or not there.

Q.3. Whether any changes in ICS models to be used under current project than ICS distributed under other ongoing projects as DelAgua group is already having other registered projects and distributing ICS from past many years.

Ans: The technology upgrades with time and PP uses the state-of-the-art ICS technology available at the time of start of the grouped project activity, under current project the PP is distributing Burn make Kuniokoa generation 2 and S32 models which are having more than 50% thermal efficiency.

There were no negative comment/concern raised by stakeholders attended the meeting.

Both the LSC meetings were recorded using digital methods (audio-visual, photographic, and pdf records of the Q&A along with attendance), and paper format of the meeting forms. The invites are also shared with the local stakeholders a month before the meeting date. The evidences are shared with the VVB for their validation.

Procedures for Ongoing communication with the local stakeholders:

Following are the channels through which the stakeholders can communicate with DelAgua.

1. DelAgua Call Center Phone Number: Aiding the local people in regional language Kinyarwanda, international languages English & French. This can be evidenced through the distribution poster shared with the end user. The distribution poster includes the call center phone number and is informed to the end user.

2. DelAgua email id: DelAgua email is another source of connection through which the end users can connect with them.

3. Through CHWs: The Community Health Workers are the point of contact for the end users in case they have to communicate with DelAgua. They rely on them, mainly because of the personal connect they have with them. CHWs are generally among the local villagers themselves, and therefore, it is easier for them to connect with the end users.

Apart from the mechanism available for local stakeholders connecting with the DelAgua, the follow-up visits by the DelAgua staff are also undertaken to user households.

During the current monitoring period, 3 feedback/grievances were raised. All three issues were resolved. Summary of the issues are provided below:

Date Raised	Raised by	Mechanism raised via.	Low / High risk	DelAgua Response to Issue raised	Date response given
15-10-2022	*HH - Rwamagana; Fumbwe*	*CHW*	*Low*	*Current Project design only allows the distribution of 1 ICS per household. DelAgua was not able to feedback due to the anonymous nature of the feedback.*	*15-10-2022*
23-01-2023	*HH - Kirehe; Mushikiri*	*CHW*	*Low*	*Current Project design only allows the distribution of 1 ICS per household. DelAgua was not able to feedback due to the anonymous nature of the feedback.*	*23-01-2023*
25-01-2023	*CHW - Burera; Cyanika*	*DelAgua Staff*	*High*	*DelAgua staff went to the market and found the stove. The stove barcode was identified and the household that the stove belongs to was visited by DelAgua staff members to understand why their stove was for sale in a local market. The HH informed DelAgua that their stove had recently been stolen and they were thankful it had been found*	*01-02-2023*

				in a market. DelAgua returned the stove to the appropriate household. The Police were informed. To date, no knowledge of any police action being taken.	

Details of the feedback/grievance has been submitted to VVB.

2.3 Environmental Impact

The project undertakes the distribution of improved cook stoves. As per the Laws of Rwanda, the project undertaking such activities does not require any Environmental Impact assessment. This is as per Ministerial Order No. 004/2008: Ministerial Order establishing the list of works, activities and projects that have to undertake an environment impact assessment.[9]

2.4 Public Comments

There were no comments received during public comment period from 17 March 2023 to 16 April 2023 on VERRA website.

2.5 AFOLU-Specific Safeguards

This Grouped Project does not allow for AFOLU project activity instances, and therefore, this section is not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The Grouped Project activity will involve PAIs that will apply the following methodology, approved under VERRA:

- VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves – version 1.1[10]

[9] https://www.elaw.org/rwanda-environmental-impact-assessment-law-and-regulations;

Specific URL to the Order No. 004/2008:
https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Felaw.org%2Fsystem%2Ffiles%2Fattachments%2Fpublicresource%2FMINISTERIAL%2520ORDER%2520N%25C2%25B0004%2520ESTABLISHING%2520THE%2520LIST%2520OF%2520WORKS.docx&wdOrigin=BROWSELINK

[10] https://verra.org/methodologies/vmr0006-methodology-for-installation-of-high-efficiency-firewood-cookstoves/

In the context of applied methodology, the associated tools and guideline documents applicable for the Grouped Project include:

- CDM TOOL30 “Calculation of the fraction of non-renewable biomass” Version 04[11];
- CDM Guideline “Sampling and surveys of CDM project activities and programmes of activities” version 04[12];
- CDM Standard “Sampling and surveys for CDM project activities and programmes of activities” version 09[13].

3.2 Applicability of Methodology

The applicability criteria of the methodologies and tools used are justified at the project activity level. This is reflected in the table below.

Table 5: Fulfilment of VMR0006, ver. 1.1 Applicability criteria for PAIs to be included

S. No.	Applicability Condition	Justification of applicability
1	Project activities shall be implemented in domestic premises or in community-based kitchens.	Since each PAI distributes technology that is appropriate for domestic use to household end users i.e. improved cookstoves for cooking, thus this meets the applicability
2	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer’s specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	ICS distributed in each PAI has a thermal efficiency of at least 25% Baseline and project fuel for each PAI has been recorded to ensure that wood fuel is the sole source of energy.
3	Both ‘Projects’ and ‘Large Projects’ can use the methodology	As per estimated annual emission reduction potential, the grouped project activity falls under large projects. As such, PAIs also fall under the Large Project category.

[11] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf

[12] https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

[13] https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

4	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	Statistics from the Forest department show that forests were estimated to cover 659,000 ha in 1960, which has reduced to 240,746 ha in 2007[14]. This reflects a loss of approximately 64 per cent of forests in between 1960 and 2007, which is more than 1.3 per cent per year. The Wood remains the main source of domestic energy for more than 90 per cent of Rwandans, where in firewood in mainly used by rural areas. The above can also be confirmed from Forest Resource Assessment report 2020 for Rwanda, which states around 45.27% loss in forest from 1984 to 2015[15] Further the study conducted in June 2009 in various districts of Rwanda reveals that 98% of rural households uses wood fuels as their main source of energy for cooking, which consist 92% firewood, 5% charcoal and 3% other residue[16]. A recent study conducted by Ministry of Land and Forestry, Rwanda in 2017[17], reveals that wood demand and supply ratio is 2:1 and the shortage is projected to increase until in future unless alternative sources of wood energy are sought. The consumption of fuelwood for Rwandan households is estimated at 2.7 million tonnes per year. The business as usual scenario on wood supply/demand, estimates the deficit between wood supply and demand to be 4.3 million tonnes (oven dry weight) in 2017, which is

[14] https://www.rema.gov.rw/soe/chap6.pdf
[15] FRA 2020 Étude de bureau, Rwanda
[16] https://www.cleancookingalliance.org/binary-data/RESOURCE/file/000/000/36-1.pdf
[17] https://www.climateinvestmentfunds.org/sites/cif_enc/files/fip_final_rwanda.pdf

		projected to increase to 7.5million tonnes by 2026. This is due to a high increase demand for fire wood and wood for charcoal. This must imply over-exploitation of already low stocked forests. The analysis also shows that 93% of total wood demand mentioned above is for cooking purpose (57% firewood and 36% charcoal). It can therefore be concluded that non-renewable biomass has been used in Rwanda since long back before 31st December 1989.
5	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Not Applicable. Biomass residues are not processed as a fuel to be used in the GP. The PAIs uses non-renewable biomass (firewood) as source of fuel for the improved cookstove. No processed biomass residue is used.
6	The Project Design Document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo)	In order to avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient Each PAI has a system for avoiding double counting with data being collected via a smartphone app (where possible), or via paper forms, including:

		1. Date of distribution 2. A unique serial number on each technology Available end-user data: name, phone number, address/physical location, ID number
7	The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The End User Agreement confirms that the ownership rights of the ER/VCUs generated by the PAI reside with the PP. The signature of each end user confirming their understanding of this agreement has been collected at the point of distribution. The End User Agreement is in English and translated into a local language where required. If necessary, it is read out to the End User. All rights to distribute the delivered stoves and claim the resultant emissions reductions belong to the PP. The GP does not include any PAI that has been rejected by another GP, or PA-DDs of any other GHG programs.

Tool 30 - Calculation of the fraction of non-renewable biomass

Applicability criterion	How the project complies
This tool may be used by: (a) DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or (b) project participants to calculate project- or PoA-specific fNRB values.	Option (b) applied to calculate the project specific f_{NRB} value as described under in the separate f_{NRB} spreadsheet submitted to VVB.

Project participants and DNAs shall identify and clearly delineate the applicable area for which fNRB is determined. For project- or PoA-specific fNRB values, project participants shall assess and use the area from which woody biomass is sourced for end-users included in the project activity and justify the selection of the area in CDM project design documents. For region- or country-specific fNRB values, DNAs shall specify the applicable area.	The PP has chosen host country Rwanda as region for the project activity.
Project participants and DNAs may choose between the following two options to determine the value of fNRB: (a) Use the default value as provided in TOOL33; or (b) Calculate fNRB by determining the share of renewable and non-renewable woody biomass in the total quantity of woody biomass consumption for the country/region or the project area (hereinafter referred as the applicable area) following the procedure and requirements in the paragraphs below. The project participants shall compare and analyse the calculated values against the values for fNRB reported in relevant scientific literature and justify any differences. This analysis shall be included in the appropriate section of the PDD. The relevant scientific literature should include at least: (i) Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. Nature Climate Change, 5(3), pp. 266–272.	The PP has chosen option b), wherein a third-party consultant has established the f_{NRB} for host country. Please refer Appendix-1 for calculation details and comparison with literature.
If the fNRB value is estimated at the national level, as a cross check, project proponent shall compare the value of estimated NRB with the product of: i) total average above ground biomass tonnage of the area of forest areas deforested in recent past (tonnes/ha), and ii) most recent available observed annual rate of deforestation (ha/yr). If the estimated NRB value is more than 10% above the value	Please refer Appendix-1 for cross check.

calculated as per the product of biomass and deforestation rate, justification shall be provided for the higher value for NRB	

3.3 Project Boundary

The Grouped Project boundary is defined as per the applied methodology VMR-0006, ver. 1.1.

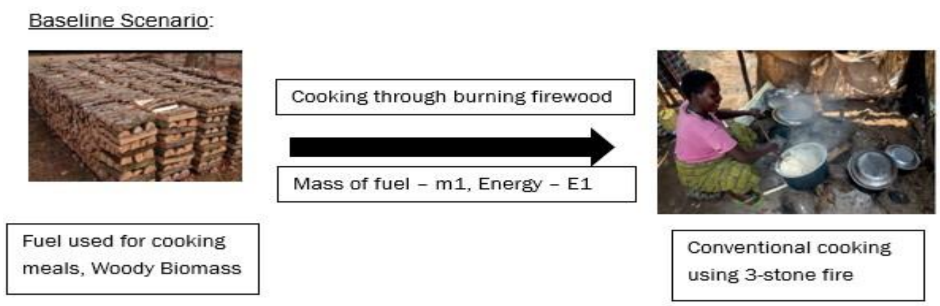
Table 7: The GHG sources & sinks within the project boundary

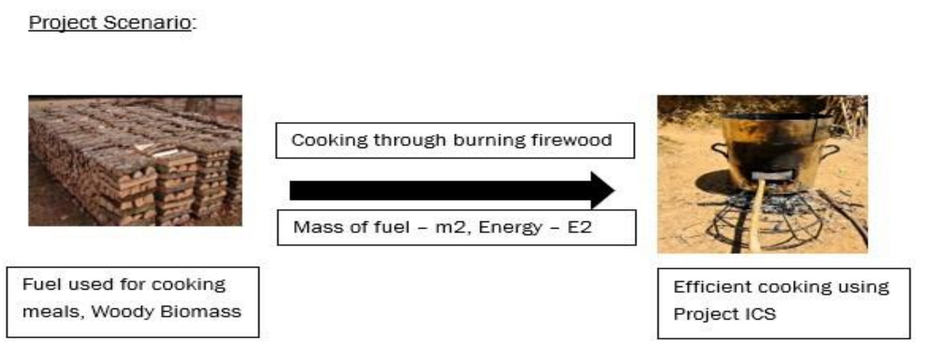
Source		Gas	Included?	Justification/Explanation
Baseline	Source 1 – for ICS (Emission from burning of non-renewable Biomass, i.e. Woody Biomass for household cooking requirements)	CO_2	Yes	Major source
		CH_4	Yes	Major source
		N_2O	Yes	Major source
		Other	No	No other source identified
Project	Source 1 – for ICS (Emission from burning of non-renewable Biomass, i.e. Woody Biomass for household cooking requirements)	CO_2	Yes	Major source
		CH_4	Yes	Major source
		N_2O	Yes	Major source
		Other	No	Not applicable

Figure 4: Project boundary map showing physical locations of the Grouped Project distribution

As per the template requirements, the diagram of the equipments distributed under the grouped project is provided in Section 1.

Flow of mass and energy within the grouped project activity is described below.





Here, the change in mass of firewood between baseline and project scenario is m1>m2.

The change in energy consumed for cooking between baseline and project scenario is E1<E2. The systems that were applied for the distribution activity to take place, are described in the Section 5.3 of this document.

3.4 Baseline Scenario

The baseline scenario is the continued use of traditional cookstove such as three stone-fired cookstove using non-renewable wood fuel (firewood) by the target population in rural and semi urban areas to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. According to the applied methodology on page 19 "A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney". The baseline three-stone cookstove has an efficiency of 10%[18].

PP conducted baseline survey and baseline KPTs between September to December 2021, to assess the type & quantity of fuels consumed and the cooking technologies in use in pre-project scenario. The PP initially conducted a desk-research to identify regions (districts) where use of traditional cookstoves and wood fuel is predominant. This was followed by designing a survey questionnaire to validate the baseline scenario in the selected regions. The baseline scenario was identified as the continued used of traditional cookstove (e.g. Rondereza) without chimney or grate or three-stone fired for preparation of meal. More than 90% households in the rural and semi-urban areas are relying on firewood for cooking.

Since the grouped project activity provides improved cookstoves to households where there was only firewood was used as the cooking fuel, and the cooking device was either a traditional cooking device (e.g. rondereza) or 3 stone fire, it is considering the burning of firewood, and usage of unimproved cooking devices as the baseline. Also, PP verifies this for every individual household where distribution of project ICS is taking place, that they were using unimproved cooking device using firewood as fuel prior to receiving it. This is acknowledged by the end-user in the form of a signed ICS usage practices poster where the factual information is written down. This information can be checked through the sample posters. The poster is in local language Kinyarwanda, which makes it clearly understandable for the end user.

Apart from the above, there are several documentary evidences in the form of recently published reports and papers that can support the fact that the project's baseline was usage of firewood and unimproved cooking devices.

Some of these are listed below:

1. Ministry of Infrastructure: Biomass accounts for 85% of all energy consumed[19]. Under the National strategy for transformation, the sector objective is to halve the number of households using traditional cooking technologies to achieve a sustainable balance between supply and demand of biomass through promotion of most energy efficient technologies.

[18] https://verra.org/wp-content/uploads/imported/methodologies/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf

[19] https://www.mininfra.gov.rw/digital-transformation-1-1

2. Rwanda Integrated Household Living Conditions Survey, EICV5: In rural areas, firewood remains the most common type of cooking fuel, used by 93% of the households[20].

Rwanda first Biennial Report to UNFCCC: Biomass (mostly wood fuel), is the most used fuel in households for cooking (83 % of households in 2019) and as a source of energy (86.3% in 2016/2017).

The survey focused on 4 provinces of Rwanda – Eastern, Northern, Southern and Western. Total 16 districts were randomly selected from for conducting the baseline survey and baseline KPTs. Enumerators were given adequate training in relation to data collection and quality checks by the PP.

PP used guidance provided in para 8.4-b of VMR0006 version 1.1. to conduct the baseline survey. Sample size was determined using CDM Guideline: Sampling and Survey for CDM PA and PoA version 4.0. Sample size for each province as per CDM Guideline on 95/10 precision confidence was between 30-33 per province for the baseline survey. PP has randomly selected 32-36 samples per province, totalling to 137 households. Baseline KPTs were also conducted on 137 households which is also in line with CDM guidelines.

Province	Sample Covered	Sample Requirement
Eastern	35	33
Northern	34	30
Southern	32	30
Western	36	33
Total	137	126

The average fuel consumption per capita per year was derived based on KPT result from 137 households. The total baseline fuel consumption per households per year is **4.409 tonnes/year.** The parameter was within the 95/10 confidence/precision.

Mean Consumption (kg/HH/day)	12.08 (4.409 tonnes/HH/Year)
Standard deviation	4.67
CoV (Standard Error)	39%
Precision (95/10)	**6.3%**
Result	Ok Acceptable

The below table has consolidated baseline fuel consumption values for various registered projects across Rwanda:

[20] https://www.statistics.gov.rw/publication/eicv-5-main-indicators-report-201617

Registry	Project ID	Geography	Baseline fuel consumption value (tonnes/device/year)	Year of Baseline Assessment
Verra	3654	Rwanda	6.753	2020
Verra	2380	Rwanda	7.635	2021
Gold Standard	GS 11012-11027 & GS10740-10742	Western Rwanda	3.170	2021
Gold Standard	GS 10990-11006 & GS 10826-10828	Eastern Rwanda	3.760	2020
Gold Standard	GS 7449 to GS 7454	Southern Rwanda	4.470	2019
Gold Standard	GS 3444-3451, 2891-2894, 3945,3947, 1267	Eastern Rwanda	4.971	2021
Gold Standard	GS 11205	Eastern & Western Rwanda	4.037	2021 & 2022
Gold Standard	GS 11028	Western Rwanda	3.20	2021
Gold Standard	GS 11026	Western Rwanda	3.20	2021
Gold Standard	GS 10999	Eastern Rwanda	3.763	2020

The value of fuel consumption derived using baseline KPTs by the project is conservative compared to values used by other VERRA registered projects which has conducted baseline assessment during the same time under VCS[21][22]. Additionally, as per National Survey on Cooking Fuel Energy and Technologies in Households, Commercial and Public Institutions in Rwanda, a report by Ministry of Infrastructure and Finance (Govt. of Rwanda), the consumption of traditional biomass is around 6.75 tonnes/HH/year. A detailed excel with list of registered projects and baseline fuel consumption value has also been submitted to VVB.

[21] https://registry.verra.org/app/projectDetail/VCS/3654

[22] https://registry.verra.org/app/projectDetail/VCS/2380

The projects registered under Gold Standard have values of baseline fuel ranging from 3.17 to 4.97 tonnes/HH/year however these are region specific data and not for the entire country. Additionally each of the projects have used different method for baseline assessment hence cannot be compared with the current project. PP has ensured the value of baseline fuel is conservative which is in line with section 2.2.1 of VCS Standard.

Since the grouped project activity provides improved cookstoves to households where there was only firewood was used as the cooking fuel, and the cooking device was a traditional cooking device such as three-stone fired, traditional stove without chimney etc., it is considering the burning of firewood, and usage of unimproved cooking devices as the baseline. Also, PP verifies this for every individual household where distribution of project ICS is taking place, that they were using unimproved cooking device using firewood as fuel prior to receiving it. This is acknowledged by the end-user in the form of a signed ICS usage practices poster where the information is written down. This information can be checked through the sample posters. The poster is in local language Kinyarwanda, which makes it clearly understandable for the end user.

3.5 Additionality

The applied methodology provides the two options to choose from, within the activity method, to demonstration additionality. This is elaborated below.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

The Grouped Project is a voluntary activity initiated by the project proponent. The initiative is solely dependent on VCU revenue for its sustenance and is completely financed by the project developer

Step 2: Positive List

Requirement:

The applicability conditions of this methodology represent the positive list. The project must demonstrate that it meets all of the applicability conditions of the methodology as well as the below condition. In so doing, the project is deemed as complying with the positive list.

The project meets all the applicability conditions as demonstrated in section 3.3 above.

1. Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.

2. Project activities that are implemented as part of government schemes or are supported by multilateral funds cannot be considered additional even if the stoves are distributed free of cost or at a highly subsidized rate and hence are not eligible to use this methodology.

Justification:

1. The ICSs under the project are to be distributed at zero cost to the end users and therefore the project fulfils the criteria for positive list. The project has no other source of revenue other than the carbon credits. Carbon revenue will only make viable the project activity.

2. The project activity is voluntary activity and not supported by any government scheme.

Based on the above substantiation no further demonstration required for additionality.

Therefore, the Grouped project activity is deemed additional.

3.6 Methodology Deviations

Not applicable.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The grouped project will have multiple project activity instances. The first 4 PAIs under the grouped project has started distribution from 11 October 2021 and till the end of monitoring period i.e. 28 February 2023, 398,608stoves distributed and operating successfully. Year-wise distribution number under each PAI till the end of monitoring period is shown below:

PAI	Year-2021	Year-2022	Year-2023	Total
PAI-1	-	-	24,274	24,274
PAI-2	499	249,466	64	250,029
PAI-3	-	99,412	-	99,412
PAI-4	-	24,878	15	24,893

The operation of stoves was confirmed by a survey undertaken as part of monitoring requirements of VCS PD and it was found that all the stoves are in use and there was no case of drop out identified during the survey. The results are survey are demonstrated in the ER spreadsheet and shall be furnished to VVB for verification. The first cookstove under this grouped project distributed on 11 October 2021, the cookstoves were distributed in the targeted area for the project, location was selected in consultation with the distribution partner. During the current monitoring period, only Burn Kuniokoa G2 stoves have been distributed. Each household has received 1 ICS. The first monitoring survey was undertaken in February 2023 by visiting the sampled households. Sample size was selected by using the latest version 4 of CDM guideline, "Sampling and surveys for CDM project activities and programmes of activities" version 04. The surveyors visited the houses and completed the survey.

The implementation status of the PAIs are shown below:

PAI	Date of inclusion of first ICS	Date of inclusion of last ICS
PAI-1	01-January-2023	27-February-2023
PAI-2	10-November-2021	08-February-2023
PAI-3	14-May-2022	12-September-2022
PAI-4	11-January-2022	21-February-2023

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The applied methodology VMR0006, ver. 1.1 does not account for baseline emissions separately, rather quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. See section 5.4 below.

5.2 Project Emissions

The applied methodology VMR0006, ver. 1.1 does not account for project emissions separately, rather quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. See section 5.4 below.

5.3 Leakage

Leakage shall be considered as default 0.95 in accordance with section 8.3 of VMR0006.

5.4 Estimated Net GHG Emission Reductions and Removals

Equations used for Improved Cookstove as per VMR0006

$$ERy = \sum_{i} \sum_{j} ERy,i,j \qquad \text{Equation (1)}$$

Where:

i = Indices for the situation where more than one type/model of improved cook stove is introduced to replace three-stone fire

j = Indices for the situation where there is more than one batch of improved cook stove of type i

ERy = Emission reductions during year y in t CO_2e

ERy,i,j = Emission reductions by improved cook stove of type i and batch j during year y in t CO_2e

$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \text{ x } NCV_{woodfuel} \times (EF_{wfCO2} + EF_{wfnon\text{-}CO2}) \times N_{y,i,j} \text{ x } 0.95$ *Equation (2)*

Where:

$B_{y,savings,i,j}$ = Quantity of woody biomass that is saved in tonnes per improved cook stove of type i and batch j during year y

$f_{NRB,y}$ = Fraction of woody biomass that can be established as non-renewable biomass (fNRB)[23]

$N_{woodfuel}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)

$EF_{wf\ CO2}$ = CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO2/TJ)

$EF_{wf\ non\text{-}CO2}$ = Non-CO2 emission factor for the use of wood fuel in baseline scenario (IPCC

default for wood fuel, 26.23 tCO2/TJ)

$N_{y,i,j}$ = Number of improved cook stoves of type i and batch j operating during year y

0.95 = Discount factor to account for leakage

The quantify of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cook stoves will be estimated as follows:

$$B_{y,savings,i,j} = B_{old} \times \left(1 - \frac{\eta_{old}}{\eta_{new,i,j}}\right) \qquad \text{Equation (3)}$$

Where:

B_{old} = Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove. The value of B old has been sourced from baseline field surveys.

η_{old} = Efficiency of baseline cookstove

23 fNRB sourced from Third party report for host country Rwanda

$\eta_{new,i,j}$ = Efficiency of the improved cook stove type i and batch j determined through water boiling test (WBT).

Efficiency of the improved cookstove will be updated as per equation below:

$\eta_{new,i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94$ Equation (5)

Where:

η_p = Efficiency of project stove (fraction) at the start of project activity

DF_{ny-1} = Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.

0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

If project households continue to use baseline cookstoves along with improved cookstoves, Bold shall be adjusted ex-post based on the percentage of project households found to continue such practice according to Equation 6 below. For such cases, the quantity of woody biomass saved $B_{y,savings,i.j}$ due to implementation of improved cook stoves shall be calculated using an adjusted value to account for ex-post use of baseline stoves in addition to improved cookstove.

$B_{old,adjusted} = B_{old} \times (1 - \mu_y)$ Equation (6)

Where:

$B_{old,adjusted}$ = Adjusted Bold to account the ex-post usage of firewood in baseline cookstove(s) by project households in addition to improved cookstove (in tonnes per device)

μ_y = Baseline stove usage factor to account for use of baseline cookstoves along with improved cookstoves.

<u>Sample VCU estimates (based on the calculation for the year 2022 of ICS monitoring estimates)</u>:

For the estimation of emission reduction, μ_y has been considered as 0.1. This is a monitored parameter and the value will be updated based on ex-post monitoring.

The B_{old} value is applied using the available baseline data.

Based on the above,

$B_{old,adjusted} = 4.409 * (1 – 0.1)$ tonnes/year

Or $B_{old,adjusted}$ = 3.968 tonnes/year

The project stove efficiency of Burn Kuniokoa G2 with pot skirt is 51.3%. The efficiency of stove without pot skirt is 43.4%. As per equation 5 of applied methodology, the efficiency by end of Year 2022 is as follows:

Year 2022 efficiency of project stove with pot skirt- 47.74%

Year 2022 efficiency of project stove without pot skirt- 40.39%

In practice it is assumed that 15% user may not use pot skirt due to various reasons/occasions, hence weighted average efficiency for year 2022 is calculated as below.

Project stove efficiency in year 2022 = 47.74% x 0.85 + 40.39% x 0.15 = 46,64%

$B_{y,savings,i,j}$ = 3.968x (1 - (0.1/0.4664) tonnes/year

Or $B_{y,savings,i,j}$ = 3.117 tonne/year

Further, it is assumed that 1% project stove might not be operational by Year-2022. Total stoves planned to be distributed is 400,000

$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \text{ x } NCV_{woodfuel} \times (EF_{wf\ CO2} + EF_{wf\ non\text{-}CO2}) \times N_{y,i,j} \text{ x } 0.95$

$ER_{y,i,j}$ = 3.117 x 0.8872[24] x 0.0156 x (112+26.23) x (400000*99%) x 0.95

$ER_{y,i,j}$ = 2,243,669 ton CO_2/annum.

Detailed calculations on the VCUs, for both (i) estimates throughout the crediting period, and (ii) actual VCUs obtained in the first monitoring period, are provided in the ER Sheet.

Emission Reduction Estimate for Improved Cookstove[25]

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
11 Oct 2021- 31 Dec 2021	379,873	0	0	379,873
01 Jan 2022-31 Dec-2022	2,243,669	0	0	2,243,669
01 Jan 2023-31 Dec-2023	2,160,906	0	0	2,160,906
01 Jan 2024-31 Dec-2024	2,082,325	0	0	2,082,325
01 Jan 2025-31 Dec-2025	2,044,721	0	0	2,044,721
01 Jan 2026-31 Dec-2026	2,033,761	0	0	2,033,761
01 Jan 2027-31 Dec-2027	2,001,708	0	0	2,001,708
01 Jan 2028-10 Oct-2028	1,607,694			1,607,694
Total	**14,554,657**	0	0	**14,554,657**

[24]fNRB as per third party report for host country Rwanda submitted to VVB

[25] For the crediting period estimates, due consideration was given to the discount factors, thus using realistic figures for the project scenario

The VCS methodology, VMR0006 is applicable to both 'Projects' and 'Large Projects'. Hence there are no limits on volume of credits from Improved Cookstove component that can be certified per annum.

6 MONITORING

6.1 Data and Parameters Available at Validation

Data/parameter fixed ex-ante for Improved Cookstove instances

Data / Parameter	B_{old}
Data unit	tons/annum
Description	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices
Source of data	Baseline KPTs, conducted according to KPT Version 4.0
Value applied:	4.4092[26]
Justification of choice of data or description of measurement methods and procedures applied	This parameter has been fixed ex-ante. The survey revealed that the average wood consumption in the Rwanda 4.409 tonnes/device/year. The sampling was done in line with para 8.4-b of applied methodology using CDM guideline: Sampling and Survey for CDM project activity and programme of activity, version 4.0. Quantitative measurement of fuel wood which is the primary fuel, done with a weighing scale and moisture meter and its consumption was measured for 3 consecutive days in the sampled households, visiting those households on 4 consecutive days. The details of the baseline KPTs including target population, sample size, sampling method, precision/confidence achieved has been added in section 3.4.
Purpose of Data	Calculation of emission reductions

[26] A baseline firewood consumption of 12.08 kg/HH/day converted into tons/hh/annum. Calculations may be checked in the ER Sheet.

Comments	B_{old} will remain fixed for the entire crediting period. The PP conducted a baseline KPT in the months of September to December 2021, to assess quantity of fuels consumed and the cooking technologies in use in pre-project scenario. The survey focused on 4 provinces (16 districts) in Rwanda. Baseline KPTs were done in 137 households to determine fuel consumption per day. The PP compared the results from the surveys with peer-reviewed publications and other registered projects and found values to be comparable/conservative.

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity
Source of data	Certification by the Centre for Research in Energy and Energy Conservation (CREEC) lab. CREEC is a certifying agent recognized by the Rwandan Standards Bureau (RSB) [27]..
Value applied:	1) Kuniokoa Generation 2: • 0.434 for project stove without skirting • 0.513 for project stove with pot skirting 2) SSM S32X: • 0.457 for project stove without skirting • 0.501 for project stove with pot skirting
Justification of choice of data or description of measurement methods and procedures applied	This data is fixed ex-ante as per applied methodology VMR0006.
Purpose of Data	Calculation of emission reductions
Comments	If a pot-skirt is distributed with the project stove, the parameter will be calculated as a weighted average of the efficiency of a skirted stove and an un-skirted stove.

[27] Certification report provided to the VVB

Data / Parameter	$NCV_{biomass}$
Data unit	TJ/ton
Description	Net calorific value of the non-renewable woody biomass
Source of data	IPCC
Value applied:	0.0156 TJ/tonne
Justification of choice of data or description of measurement methods and procedures applied	Default value as per VMR0006, V1.1
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	$EF_{wf\ CO2}$
Data unit	ton CO2/TJ
Description	CO_2 emission factor for woody biomass
Source of data	IPCC
Value applied:	112 tCO2/TJ
Justification of choice of data or description of measurement methods and procedures applied	Default value as per VMR0006, V1.1
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	$EF_{wf\ non\text{-}CO2}$

Data unit	ton CO2/TJ
Description	Non-CO_2 emission factor for woody biomass
Source of data	IPCC
Value applied:	26.23 tCO2/TJ
Justification of choice of data or description of measurement methods and procedures applied	Default value as per VMR0006, V1.1
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	Leakage Factor
Data unit	Fraction
Description	net to gross adjustment factor to account for the leakage
Source of data	Default as per VMR0006, ver. 1.1, Section 8.3
Value applied:	0.95
Justification of choice of data or description of measurement methods and procedures applied	In-line with the applied methodology VMR0006, ver. 1.1 Section 8.3.
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction

Description	Fraction of woody biomass that can be established as non-renewable biomass
Source of data	f_{NRB} report prepared by C4 EcoSolutions (Pty) Ltd using CDM Tool 30 Version-04.0
Value applied	0.8872
Justification of choice of data or description of measurement methods and procedures applied	The applied value is calculated by applying the latest available version of CDM Tool-30 (version 04.0) by third party.
Purpose of Data	Calculation of emission reductions
Comments	The applied f_{NRB} value has been calculated by using the latest available data sources. As per the fNRB calculation done by PP, the value is 91% however as a measure of conservativeness, PP used the fNRB value from another registered projects which was lesser than 91%. Hence, the value has been changed to 88.72%.

6.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Source of data	Monitoring surveys
Description of measurement methods and procedures applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve95/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and program of activities
Frequency of monitoring/recording	At least once every two years
Value applied	400,000 (for ex-ante estimation)
Monitoring equipment	Monitoring / Usage Surveys

QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	N/A
Comments	This parameter only accounts for the number of project ICS that are operating in the monitoring period, and rest of the factors such as discount on non-usage of project ICS, usage of pre-project device, and drop-off are accounted separately, and hence are not reflected with the values provided for this parameter. Here, the batch j is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline stove
Source of data	Calculated using the Default value as specified in VMR0006 methodology and the data obtained through the distribution surveys
Description of measurement methods and procedures applied	If the device is a three stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney, a default value of 0.1 is used as per the applied methodology; for other types of devices, a default value of 0.2 is used.
Frequency of monitoring/recording	Fixed for each individual household at first HH visit
Value applied:	0.10 for ex-ante only
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions.

Calculation method	N/A
Comments	The baseline stove is traditional stove such as three stone fired stone using firewood. Hence, efficiency of stove is used 10% which is VMR0006 default. In case there is any other unimproved cooking device identified as baseline cooking device, then the applied value for the parameter will be 0.2. The average of the ηold recorded for total project ICS households (in the distribution surveys) will be calculated for VCU calculations. For ex-ante estimates, the applied value is 0.1 (**assumption for the ex-ante calculations**), which will vary for each monitoring period, based on the actual data obtained at the time of distribution.

Data / Parameter	$\eta_{new,i,j}$
Data unit	Fraction
Description	Efficiency of the device of each type i and batch j
Source of data	Calculated as per the equation 5 of the applied methodology
Description of measurement methods and procedures applied	Efficiency of the improved cookstoves will be estimated each year using equation (5) above where loss in efficiency per year is calculated, and therefore no direct measurement is involved.
Frequency of monitoring/recording	Annually
Value applied:	Kuniokoa ICS efficiency value for the crediting period years:

Crediting Period Year	Kuniokoa Efficiency value		SSM S-32 X	
	With pot skirt	Without pot skirt	With pot skirt	Without pot skirt
Year-1	0.4822	0.4080	0.4709	0.4296
Year-2	0.4774	0.4039	0.4662	0.4253
Year-3	0.4726	0.3998	0.4616	0.4210

	Year-4	0.4679	0.3958	0.4570	0.4168
	Year-5	0.4632	0.3919	0.4524	0.4127
	Year-6	0.4586	0.3880	0.4479	0.4085
	Year-7	0.4540	0.3841	0.4434	0.4044
	Year-8	0.4495	0.3802	0.4389	0.4004
Monitoring equipment	N/A				
QA/QC procedures applied	The values that will be applied will be based on the model of ICS distributed.				
Purpose of data	Calculation of emission reductions				
Calculation method	As per equation (6) above				
Comments	Different models of ICS may be used at the time of distribution (hence monitoring), as indicated in Section 1.11 of PD&MR. Also, the batch j is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.				

Data / Parameter	μ_y
Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y
Source of data	Monitoring
Description of measurement methods and procedures applied	This parameter should be monitored using one of the following methods: If the baseline cookstoves are decommissioned and no longer used, as determined by the monitoring survey its value is 0 and $B_{old, adjusted}$ is equal to B_{old}. If both the improved cookstove and baseline cookstoves are used together then surveys shall be conducted to record the average continued operation of baseline cookstoves in a sample of households. The surveys should be designed to capture the cooking habits and stove usage of households in the region, including quantification of use of baseline cookstoves, by formulating questions and/or collecting

	evidences to determine the frequency of usage of both the improved cookstoves and baseline cookstoves.
Frequency of monitoring/recording	At least once every two years
Value applied:	0.10 (assumed for ex-ante estimation)[28]
Monitoring equipment	Monitoring /Project Survey
QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions
Calculation method	N/A
Comments	N/A

Data / Parameter	**Life span**
Data unit	Years
Description	State the operating lifetime of project device for projects opting Equation 6 (above) for updating project stove efficiency during project crediting period.
Source of data	Manufacturer's specifications
Description of measurement methods and procedures applied	The operational life of cookstove is sourced from manufacturer specification.
Frequency of monitoring/recording	Yearly
Value applied:	1) Burn Kuniokoa Generation 2 = 10 years 2) SSM S32X = 10 years
Monitoring equipment	N/A

[28] Applied as assumption for year-1 estimations.

QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions
Calculation method	N/A
Comments	The emission reduction from each project stove can be claimed till 10 years operational lifetime of the project stove from date of commissioning, for onward if PP willing to claim emission reduction PP may replace project stove with new improved cookstoves subject to baseline applicability at the time of replacement.

Data / Parameter	**Date of commissioning of batch j**
Data unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in "batches" and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch.
Source of data	Distribution database
Description of measurement methods and procedures applied	To be recorded once at the time of distribution for each project activity instance
Frequency of monitoring/recording	Once
Value applied:	11-October-2021 (date when the first ICS under the PAI is commissioned)
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions
Calculation method	N/A

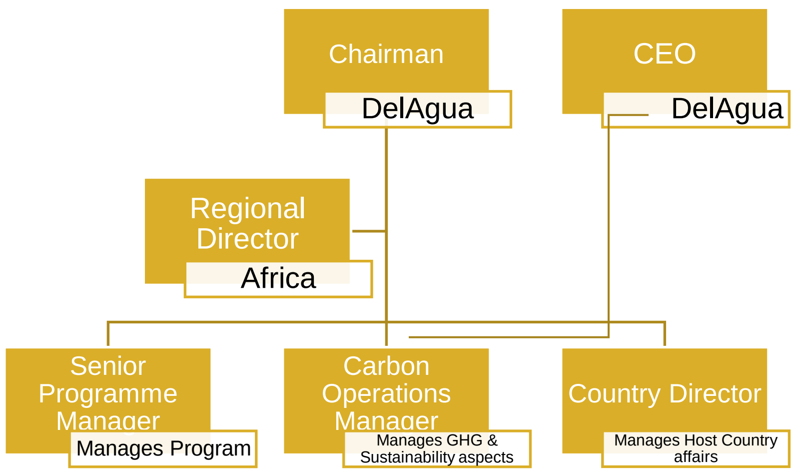
Comments	N/A

6.3 Monitoring Plan

The VCS Grouped Project in Rwanda will be following VCS standard requirements applicable, along with the methodological requirements on monitoring a grouped non-AFOLU large project activity. The Monitoring Plan is elaborated in multiple sub-sections for the sake of clarity.

Team Involved: The DelAgua team involved in the GHG operations management is the local staff based in Rwanda along with the senior management.

The key responsibilities are shared by the team as follows:



Roles and Responsibility:

Chairman – Supervises the overall programme at strategy and financial side.

CEO – Acts as central point of strategy, programme management, finance, and technical domains of the programme.

Regional Director – Responsible for the regional level programme implementation and expansion planning and implementation.

Senior Programme Manager – Manages the programme at implementation level, and acts as an internal technical reviewer for the programme documents and procedures.

Carbon Operations Manager – Manages the operations for GHG emission mitigation programme, from planning to implementation stage. Additionally manages the Sustainability aspects of the programme.

Country Director – Supports programme in the operations and management at the host country level, and acting as point of communication between DelAgua and government of Rwanda.

The collective responsibilities of DelAgua as the Project Participant is indicated below under para 2.

The monitoring plan for this project is closely derived from the methodologies. A database for the project activity will be maintained by the DelAgua.

1. Monitoring Plan Design for the Grouped Project Activity:

The monitoring plan shall consist of checking a representative sample of all ICS units distributed at least once every two years to ensure that they are still operating or are replaced by an equivalent in-service unit. A project database will be populated during the distribution of each initial cookstove issued, and will be updated based on subsequent replacements, as well as detailed data on the representative sample surveyed for monitoring purposes. The database will be accessible to the project proponent, appropriate partners, and the verification VVB. The database will include, at minimum, the following:

- Unique project identification number
- Installation date
- Family name
- Contact details of user (when available)
- Socioeconomic level (if households or communities are targeted), or type of establishment (if SMEs are targeted)
- Baseline cooking fuel source
- Baseline stove type
- Date of replacement of cook stove
- Monitored parameters
- Usage of Skirt with the project ICS

The database will be available to select a random, representative sample for monitoring and verification purposes. This sample set will be integrated into the database to include additional monitoring parameters as required or as appropriate. Monitoring surveys will include examination of the cookstove installations and verification that they are in an operational condition. Units that have been damaged or aged beyond expected operational condition will be replaced. Therefore, should some appliances not be replaced appropriately, this will be accounted for and appropriately deducted from emission reduction claims.

2. Responsibility

The main roles and responsibilities of the parties involved is as described below:

DelAgua

- Acts as the project managing entity

- Development of PD
- Registration of the project with VERRA
- Inclusion of new PAIs upon satisfaction of the eligibility criteria stipulated in the PDD
- Authorized entity for any official communication with the VERRA and VVB
- Manage and archive the database for all appliances
- QA/QC of database during distribution & data collection
- Maintain local repair/replacement facilities
- Education and public awareness raising during initial distribution as well as periodic education campaigns.
- Data collection during distribution & delivery of data to database
- QA/QC of data collection
- Conduct ongoing monitoring, as required to ascertain monitored parameters.

3. Quality Assurance and Quality Control

Spot checks of the database are conducted during the baseline surveys, distribution, and monitoring activities. Distributors are trained in the proper installation, use, and maintenance of the devices, and are provided with training sufficient to introduce recipient households to the proper operation and maintenance of the devices, as well as contact information for DelAgua. Supervisors receive similar training, as well as supervisory training in monitoring of distributor's activities. Supervisors are responsible for approximately 10 distributors. During the campaign, DelAgua staff will spot-check distributor quality, by revisiting 2 households per distributor, to verify that the stoves are in place, and that the recipient can show proper operation of the units. The results of this spot check are recorded electronically and compared against the initial distribution logs. The data collected in the database is verified and controlled in the following way:

1. Surveys are conducted by staff trained by DelAgua.
2. Supervisors are separately trained by DelAgua to spot-check surveyor results.
3. All ex-ante and monitored data parameters are collected on smart-phone based surveys and are uniform across all surveyors for any given survey activity.
4. Unknown to the surveyors, the database analysis includes verification of "valid" surveys including by a.) verifying household consent is received and b.) that the survey duration, as recorded by Internet time, is no shorter or longer than a reasonable value. For

example, the baseline surveys for Rwanda were validated against a minimum of 20 minutes and a maximum of 120 minutes.

5. Data validation is conducted on the database to ensure that values are numerically consistent, i.e. multiple choice values add up to 100% of respondents.

4. Training and maintenance procedures:

DelAgua is responsible for training any contractors used during distribution, education and monitoring activities. DelAgua ensures training of all on-site staff with respect to adherence to the Monitoring Plan of the project activity. Records of the training will be kept. Internal audit of all the records of the distribution, monitoring and education activities are performed on an annual basis, and will be carried out once (or twice, based on the requirements and plans) a year. Training records are saved for such training programs, where step-by-step training is provided to the surveyors for a quality data collection process, where they are expected to not impact the materially, degree of conservativeness, and perform the surveys in an unbiased manner. The surveyors are trained on a periodic basis, and refresher trainings are also provided to improve the quality of surveys from time to time. During these audits all the data and parameters that need to be monitored as per the monitoring plan will be checked and shortcomings if any will be reported and addresses. Since the surveys are electronic in format, recording of responses and the record keeping procedure for these surveys are conveyed to the team through rigorous trainings.

5. Calculation of emission reductions:

Emission reductions will be calculated using the results of the most recent survey data. The updated parameters adjust all emission reduction results for the year monitored.

6. Sampling Plan:

STEP 1: Monitoring Objectives

The objective was to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of monitoring period.

a) ***Ny,I, j*** *Number of project devices of type i and batch j operating during year y (Annual/biennial)*

b) μ_y *Adjustment to account for any continued use of pre-project devices during the year y*

As per Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities version 9.0, 95/10 confidence/precision has been adopted as this is a large-scale project for both the parameters.

STEP 2: Target Population

PAI-1: 24,714

PAI-2: 250,029

PAI-3: 99,412

PAI-4: 24,895

STEP 3: Sampling Frame

The sampling frame referred to all the information sources on the database. PP collected all the information sources on monitoring data through physical visit. The monitoring survey data was collected primarily through physical survey which is the primary mechanism in place for the data collection of the newly distributed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) which is used in the current monitoring period.

Following information were recorded during the monitoring survey.

a. Product based Unique Identification number.

b. Contact information, geographical location or other relevant information of the end user

c. Sampling frame was representative of the entire population

d. No elements from outside the population of interest is present for the current monitoring period. Wherever information miss-match is observed, it is corrected in the database and conservative assumptions is applied.

STEP 4: Sample Method

The sampling method for the monitored parameters $N_{y,i,j}$, and u_y is Simple Random Sampling and samples were randomly selected from the primary sampling units as illustrated above.

To ensure a random selection of ICS, random number generators was applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame.

Applying the random number generators, the ICS was then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent.

To determine the parameters, sampling involves the following approaches

$N_{y,i,j}$: Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)

$\mu_{y,i,j}$: Pre project device only is in use then fraction to be used to calculate total number, however if pre project device is used along with project ICS, proportion of usage of each was determined by cooking habits evaluated by survey questionnaire during the monitoring period. Using the formulas below, the project proponent randomly sampled the required number of ICS from the primary sampling units

STEP 5: Desired Precision / Expected Variance and Sample Size

To estimate the number of premises to be surveyed with a 10% margin of error at desired confidence level of 95%, the optimal sample size n is given by:

1. For proportional parameter i.e. number of operational project cookstoves

$$n \geq \frac{1.96^2\ N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.96^2\ p(1-p)}$$

Here,

Parameter	Description
n	Sample Size
N	Total number of cookstoves (ICS) distributed
p	Expected proportion
1.96	Represents the 95% confidence required
0.1	Represent the 10% relative precision

2. For mean-based parameter i..e adjustment factor

$$n \geq \frac{1.96^2\ NV}{(N-1) \times 0.1^2 + 1.96^2\ V}$$

Here,

Parameter	Description
n	Sample Size
N	Total number of cookstoves (ICS) distributed
SD	Standard deviation
V	$(SD/mean)^2$
1.96	Represents the 95% confidence required
0.1	Represent the 10% relative precision

Requirements of Guideline for Sampling and surveys for CDM project activities and program of activities, version 4.0 has been followed.

STEP 6: Minimum Sample Size[29]

- <u>Proportion based parameter ($N_{y,i,j}$)</u>

As per the CDM guideline for Sampling and survey[30], if the sample size calculation returns a value of less than 30 samples, a minimum sample size of 30 shall be chosen. The sample size calculation for PAI-1 has been shown below:

Sampling Constants	Values
Monitoring period start	11-10-2021
Monitoring period end	28-02-2023
Level of Sampling	Single vintage
Confidence (%) (90 or 95)	95%
Margin of Error (%)	10%
Z value	1.960

Usage Rate ($N_{y,i,j}$)				
Year	Stove Model (Sampling Frame)	Stove Population (sampling frame size)	Usage Rate ($N_{y,i,j}$) (%)	Calculated Sample Size (n)
2022	Kuniokoa	24,274	90%	52
Sample size determination				
Estimated $N_{y,i,j}$ (p)				80.00%
$V_{ny,i,j} = p(1-p)$				0.160
Sample Size required ($N_{y,i,j}$)				43
Monitoring Results				
Year	Stove Model (Sampling Frame)	Sampling frame size	Monitored Sample Size	Monitored Usage
2022	Kuniokoa	24,274	120	100%
Reliability Check				
Samples Monitored				120
Monitored $N_{y,i,j}$ (p)				100.00%
Standard Error of $N_{y,i,j}$				0.00%
Relative precision (Margin of error)				0.00%
Result				Ok, acceptable
Lower Bound confidence value				not applicable

As per the sample size calculation, 43 samples were calculated to be surveyed, However, 120 samples were considered to compensate for any outliers or non-responses associated with samples and to get accurate results. The parameter was within the 95/10 confidence/precision.

- <u>Mean based parameter (u_y)</u>

[29] Detailed sample size calculation of PAI-2, PAI-3 and PAI-4 can be found in the ER Calculation sheet

[30] https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

If the parameter of interest is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be used if the resulting sample size is less than 30. As per the CDM: Sampling and Surveys for project activities and programme of activities, PP used t-distribution to calculate the sample size.

Use of baseline stove (μ_y)					
Year	Stove Model (Sampling Frame)	Stove Population (sampling frame size)	Expected Mean value (tonnes per year)	Expected SD	Calculated Sample Size (n)
2022	Kuniokoa	24,274	0.1	0.01	7
Sample size determination					
Estimated (p)					0.10
Estimated Standard Deviation (SD)					0.01
$V_{mean} = (SD/p)^2$					0.01
Minimum Sample Size required					4
tDistribution sample size adjustment				Iteration 1	11
				Iteration 2	5
				Iteration 3	8
				Iteration 4	6
				Iteration 5	7
				Iteration 6	6
Monitoring Results					
Year	Stove Model (Sampling Frame)	Sampling frame size	Monitored Sample Size	Monitored Mean Value of use of baseline stove Monitored Standard Deviation	
2022	Kuniokoa	24,274	120	0.0167	
Reliability Check					
Samples Monitored(p)					120
Standard error of μy					1.17%%
Relative precision (Margin of error) (%)					1.14%
Result					ok acceptable
Applicable Value					not applicable

As per the sample size calculation, 7 samples were calculated to be surveyed, However, 120 samples were considered as per CDM sampling standard. There were two reasons- firstly, common survey was conducted for $N_{y,i,j}$ and μ_y and secondly, to compensate for any outliers or non-responses associated with samples. The parameter was within the 95/10 confidence/precision.

6. Quality Assurance/Quality Control

The project proponent applied a confidence/precision of 95/10 for each monitored parameter allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to both parameters i.e. number of operational cookstoves, and adjustment factors. The project proponent monitored a particular parameter till the required level of confidence/precision has been reached, as long as the calculated minimum number of samples has been achieved.

The monitoring plan followed procedures in place to ensure good quality data. The project proponent ensured that field personnel have reviewed, understood and have agreed to follow the monitoring plan procedures, including provisions for maximizing response rates, documenting out-of-population cases, refusals and other sources of non-response. A quality control and assurance strategy were audited and documented. The monitoring plan explained how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household is chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires was developed. The calculation of the sample size was carried out using values from pilot study for parameter proportions (for number of operational cookstoves), mean values (for adjustment factor) and standard deviations, as the actual characteristics of the population/sampling frame are unknown.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Value applied:	PAI-1 (BAC ICS): 24,274 PAI-2 (ABC ICS): 250,029 PAI-3 (DA ICS): 99,412 PAI-4 (ST ICS): 24,893
Comments	The project implementor maintains a database for distribution record to calculate this parameter. The sampling was undertaken by complying the 95/10 confidence precession as survey sample conducted for 4 PAIs separately. As per survey results all PAIs have achieved the required confidence and precision level. All the ICS units distributed during M.P.-1 are considered for the VCU calculation, as the monitoring surveys provide a 100% operational rate (i.e. there is no drop-off observed during M.P.-1).

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline stove
Value applied:	0.1
Comments	A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood, or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney

Data / Parameter	$\eta_{new,i,j}$
Data unit	Fraction
Description	Efficiency of the device of each type i and batch j
Value applied:	<table><tr><td></td><td>2021</td><td>2022</td><td>2023</td></tr><tr><td>PAI-1</td><td>-</td><td>-</td><td>0.4789</td></tr><tr><td>PAI-2</td><td>0.4746</td><td>0.4746</td><td>0.4651</td></tr><tr><td>PAI-3</td><td>-</td><td>0.4767</td><td>0.4767</td></tr><tr><td>PAI-4</td><td></td><td>0.4738</td><td>0.4644</td></tr></table>The pot skirt usage rate for PAI-1 is 95.54%, PAI-2 is 89.71%, PAI-3 is 92.52% and PAI-4 is 88.71%.
Comments	PP adopted Option V given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 5 of applied approved methodology where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”. For capturing the pot-skirt usage, project survey was conducted.

Data / Parameter	**Life span**

Data unit	Years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Value applied:	Burn Kuniokoa G2 – 10 years
Comments	The operational lifetime of the ICS considered as per manufacturer specification. Other models of firewood stoves in future instances shall use the manufacturer's life time for respective models.

Data / Parameter	Date of commissioning of batch j
Data unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in "batches" and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch.
Value applied:	11-October-2021 (date when the first ICS under the PAI is commissioned)
Comments	As per manufacturer specification. During the current monitoring period, only Burn Kuniokoa has been distributed.

Data / Parameter	μ_y
Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y
Value applied:	PAI-1 (AC ICS): 0.0167 PAI-2 (BC ICS): 0.0440 PAI-3 (DA ICS): 0.0240

	PAI-4 (ST ICS) | 0.0390
Comments	A 95 /10 confidence / margin of error has been achieved for the sampling parameter irrespective of annual/ biennial monitoring frequency as per para 22 of Standard: Sampling and surveys for CDM project activities and programmes of activities, Version 09.0. The desired precision/confidence was met for the parameter during the current monitoring period. Total sample monitored for this parameter for PAI-1, PAI-2, PAI-3, PAI-4 is 121, 122, 121, 118 respectively as PP conducted common survey for parameter $N_{y.i.j}$ and μ_y.

7.2 Baseline Emissions

Not applicable. The applied methodology VMR0006 does not have provision to calculate baseline emissions as separate.

7.3 Project Emissions

Not applicable. The applied methodology VMR0006 does not have provision to calculate project emissions as separate.

7.4 Leakage

A net to gross adjustment factor of 0.95 is already part of the methodological equation to calculate the Emission Reductions. The methodology states that there is no requirement to monitor leakage as separate if this net to gross adjustment factor is applied for the ER calculations.

7.5 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology was applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j}$$ Equation (1)

Where:

i	=	Indices for the situation where more than one type/model of improved cookstove is introduced to replace three-stone fire
j	=	Indices for the situation where there is more than one batch of improved cookstove of type i

ER_y	=	Emission reductions during year y in t CO_2e
$ER_{y,i,j}$	=	Emission reductions by improved cookstove of type i and batch j during year y in t CO_2e

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times \left(EF_{wf,CO2} + EF_{wf,non\ CO2}\right) \times N_{y,i,j} \times 0.95 \qquad \text{Equation (2)}$$

Where:

$B_{y,savings,i,j}$	=	Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j during year y
$f_{NRB,y}$	=	Fraction of woody biomass that can be established as non-renewable biomass (f_{NRB})
$NCV_{wood\ fuel}$	=	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)[31]
$EF_{wf,CO2}$	=	CO_2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO_2/TJ)[32]
$EF_{wf,non\ CO2}$	=	Non-CO_2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO_2/TJ)[33]
$N_{y,i,j}$	=	Number of improved cookstoves of type i and batch j operating during year y
0.95	=	Discount factor to account for leakage

To calculate $B_{y,savings,i,j}$ we use Option 1 of the applied methodology[34]

$$B_{y,savings,i,j} = B_{old} \times \left(1 - \frac{\eta_{old}}{\eta_{new,i,j}}\right)$$

Where:

B_{old} = Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove. The value of B_{old} has been sourced from baseline field surveys.

η_{old} = Efficiency of baseline cookstove

[31] 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction

[32] 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

[33] 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

[34] Equation 3 of methodology VMR 0006

$\eta_{new,i,j}$ = Efficiency of the improved cook stove type i and batch j determined through water boiling test (WBT).

Efficiency of the improved cookstove will be updated as per equation 5 of applied methodology as below:

$$\eta_{new,i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Where:

η_p = Efficiency of project stove (fraction) at the start of project activity

DF_n^{y-1} = Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.

0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

As some of the project households continue to use baseline cookstoves along with improved cookstoves, B_{old} has been adjusted ex-post based on the percentage of project households found to continue such practice according to the Equation 6 of applied methodology as follows:

$$B_{old,adjusted} = B_{old} \times (1 - \mu_y)$$

Where:

$B_{old,adjusted}$	=	Adjusted B_{old} to account the ex-post usage of firewood in baseline cookstove(s) by project households in addition to improved cookstove (in tonnes per device)
μ_y	=	Baseline stove usage factor to account for use of baseline cookstoves along with improved cookstoves.

For example, for PAI-2, considering one ICS/household, the saving for the entire monitoring is calculated as:

Year-2021

$B_{y,savings,i,j} = B_{old\ adjusted} * (1- \eta_{old}/ \eta_{new,i,j})$

$= B_{old} * (1-\mu_y) * (1- \eta_{old}/ \eta_{new,i,j})$

$B_{y,savings,i,j}$= 4.409* (1 - 0.0440) X [1- (0.10/0.4746[35])]

= 3.33 tonnes

$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times (EF_{wf,CO2} + EF_{wf,non\ CO2}) \times N_{y,i,j} \times 0.95$

= 3.33 x 0.8872 x 0.0156 x (112+26.23) * (1*100%) x 0.14* 0.95

[35] Refer to cell C38 under "Data & Parameter" worksheet in the ER calculation sheet

= 0.845 tCO_2e

Total ERs for 499 stoves in 2021 in PAI-2 is

Year	$n_{new,i,j}$ (efficiency of cookstove)	$B_{y,savings,i,j}$ (Quantity of wood saved)	$N_{y,i,j}$	$ER_{,i,j}$ (Emission reduction in tCO_2)
1	47,46%	3.33 Tonnes	499	421

Year-2022

$B_{y,savings,i,j} = B_{old\ adjusted} * (1- \eta_{old} / \eta_{new,i,j})$

$= B_{old} * (1-\mu_y) * (1- \eta_{old} / \eta_{new,i,j})$

$B_{y,savings,i,j}$= 4.409* (1 - 0.0440) X [1- (0.10/0.4746[36])]

= 3.33 tonnes

Total ERs for 249,965 stoves in 2022 in PAI-2 is

Year	$n_{new,i,j}$ (efficiency of cookstove)	$B_{y,savings,i,j}$ (Quantity of wood saved)	$N_{y,i,j}$	$ER_{,i,j}$ (Emission reduction in tCO_2)
1	47,46%	3.33 Tonnes	249,965	570,699

The methodology does not account for project emissions separately, but instead quantifies net GHG emission reductions achieved by this project

Leakage is considered as default 0.95 in accordance with Section 8.3 of the VMR0006 methodology.

The VCUs obtained during M.P.-1 are provided in the table below.

The detailed calculations for this and other PAIs may be referred from the ER Sheet.

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
11-October-2021 - 31—December-2021	421	-	-	421

[36] Refer to cell C38 under "Data & Parameter" worksheet in the ER calculation sheet

01-January-2022 – 31-December-2022	989,019	-	-	989,019
01-January-2023 - 28-February-2023	378,483			378,483
Total	**1,367,923**	-	-	**1,367,923**

Ex-ante emissions reductions /removals	Achieved emissions reductions /removals	Percent difference	Justification for the difference
2,966,919	1,367,923	-53.89%	The decrease in emission reduction is mainly due to ICS distributed in phases during current monitoring period and average operational age of ICS are significantly lower than number of days in current monitoring period, which is appropriate.

APPENDIX 1: FNRB CALCULATIONS

The f_{NRB} report using CDM Tool 30 version 4.0 was prepared by a third part expert C4 EcoSolutions (Pty) Ltd. They have 16+ years of experience of working on climate change, biodiversity, ecosystem restoration and scientific research/publications. They have been working in 105 countries across Africa, Asia, Europe, Asia Pacific, Latin America and the Caribbean. They have a diverse group of people working which includes scientist, hydrologist, botanist, climate change experts, forestry experts etc. Details of the work and experience can be found on https://c4es.co.za/.

Step 1: Estimation of renewable biomass (RB) in the country of Rwanda as per equation (4) of the fNRB tool

$$RB = \sum\left(MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})\right) + \sum\left(MAI_{other,i} \times (F_{other,i} - P_{other,i})\right) \quad \text{Equation (4)}$$

Where:

$MAI_{forest,i}$	=	Mean Annual Increment of woody biomass growth per hectare in sub-category *i* of forest areas in the relevant period (tonnes/ha/yr)
$MAI_{other,i}$	=	Mean Annual Increment of woody biomass growth per hectare in sub-category *i* of other land areas in the relevant period (tonnes/ha/yr)
$F_{forest,i}$	=	Extent of forest in sub-category *i* in the relevant period (ha)
$F_{other,i}$	=	Extent of other land in sub-category *i* in the relevant period (ha)
$P_{forest,i}$	=	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in sub-category *i*) in the relevant period (ha)
$P_{other,i}$	=	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other land areas (in sub-category *i*) in the relevant period (ha)
i	=	Sub-category *i* of forest areas and other land areas[3]

Parameters used:

Input parameter	Value	Data source
Extent of forest and other wooded land in 2021 (in ha), $F_{forest,i}$	**467,239**	Geospatial data products for Rwanda were analysed to estimate Rwanda's renewable biomass. The forest and other wooded land cover for 2000 and 2021 was estimated using Hansen/UMD/Google/USGS/NASA[37] spatial data for 2000 and 2021, which is derived from Hansen
Protected areas within forest areas (in ha)	137,289	
Remote areas within forest areas (in ha)	12,681	

37 Hansen/UMD/Google/USGS/NASA. Hansen Global Forest Change v1.9 (2000-2021). Available at: https://glad.earthengine.app/view/global-forest-change. Accessed on: 31 May 2023.

		et al.[38], and disaggregated according to the FAO[39] global ecological zones. The total woody cover extent, including forests and other wooded areas, was calculated for each ecological zone, within the protected areas and within areas that are either accessible or geographically remote.
Extend of non-accessible area (in ha), $P_{forest,i}$	**149,970**	Calculated (protected forest + remote forest)
Mean Annual Increment (t/ha/year), $MAI_{forest,i}$	1.82	The mean annual increment for forest is calculated as default age weighted average of the forest type and as per IPCC[40] values for Tropical Mountain system.

Geographically remote areas were determined to be areas beyond a harvestable distance from roads and settlements. Conservatively, it was assumed that forests and wooded areas adjacent to all roads within a distance of 2.5 km are harvestable, despite how geographically remote or inaccessible these roads may be. This threshold was determined based on the peer-reviewed literature on wood harvesting practices.

Banks et al.18 found that woody biomass density increases significantly as a function of distance from the edge of a settled area. Beyond 450 m, average biomass density was found to increase from 2.1 to 16.6 t/ha. Jumbe and Angelsen[41] modelled fuelwood choice from household survey data. They found that the most significant determinants of fuelwood choice are source attributes (size and species composition) and distance to source. The mean harvesting distance for their study was found to be 0.95 km. Bandyopadhyay et al.[42] matched household survey and remote sensing data to investigate wood fuel use in relation to poverty levels. They concluded that the one-way walk duration to wood fuel source for both poor and non-poor households is 0.6 hours. Assuming an average walking speed of 4 km/hr over uneven terrain, the average one-way distance to wood fuel source is less than 2.5 km. Areas beyond the harvestable distance were, therefore, determined to be geographically remote. Accessible areas are those that are not protected and not geographically remote, from which the annual renewable biomass production is calculated.

The default age-weighted mean annual increment (MAI) estimates of each ecological zone, as reported by the IPCC, were used for this study. The proportion of forest stand ages above and below 20 years old, including primary forests, were estimated for each ecological zone by extrapolating the observed forest gain extents between 2000 and 2012 to a 21-year period. The average MAI

[38] Hansen, M. C. C. et al. High-resolution global maps of 21st-century forest cover change. Science (80- .). 342, 850–854 (2013)

[39] FAO. 2012. Global ecological zones for FAO forest reporting: 2010 Update. Forest Resources Working Paper 179. Food and Agriculture Organisation of the United Nations, Rome. Available at: https://www.fao.org/3/ap861e/ap861e00.pdf.

[40] IPCC. Forest Land. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use 4, 1–29 (2019)

[41] Jumbe, C. B. L. & Angelsen, A. Modeling choice of fuelwood source among rural households in Malawi: A multinomial probit analysis. Energy Econ. 33, 732–738 (2011).

[42] Bandyopadhyay, S., Shyamsundar, P. & Baccini, A. Forests, biomass use and poverty in Malawi. Ecol. Econ. 70, 2461–2471 (2011)

estimates for Rwanda is 1.82 t/ha/year for tropical mountain system between 2000 and 2012 to a 20 years period.

FAO Global Ecological Zone	**Total forest cover** (ha)	**Protected area cover** (ha)	**Remote area cover** (ha)	**MAI** (t/ha/yr)	**Annual growth** (t/yr), **R**
Tropical mountain system	467,239	137,289	12,681	1.82	578,495
Total	467,239	137,289	12,681	-	578,495

Total renewable biomass RB = 578,495 t/year

Step 2: Estimation of consumption of woody biomass (H) in the country of Rwanda as per equation (2) of the fNRB tool

NRB = H - RB

Where,

H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

H = HW x N + CE + NE

Where,

HW = Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes/household)

N= Number of households consuming wood fuel within the applicable area in the relevant period (number)

CE= Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

NE= Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

The total woody biomass consumption combines the country's domestic and non-domestic *energy* and non-domestic *non-energy* consumption. Rwanda's estimate of total woody biomass consumption estimation is derived from World Bank population statistics[43] and consumption data sourced from literature[44,45], the FAO[46] and KPT surveys. Reported values of charcoal consumption are converted to the equivalent woody biomass using guidelines on kiln efficiency

43 https://data.worldbank.org/.
44 United Nations & African Union. Review of woodfuel biomass production and utilization in Africa
45 FAO. Forestry Sector in 2020: https://www.fao.org/faostat/en/#data/FO
46 FAO Statistics. Forest Products 2019. https://www.fao.org/3/cb3795m/cb3795m.pdf (2021)

from the FAO forestry paper on industrial charcoal making[47] and the latest CDM Tool 30 v04.0 2022 default charcoal-to-wood biomass conversion factor. The CDM default value of 4 was applied to the urban domestic charcoal consumption and non-domestic energy charcoal consumption.

Data was drawn from the most recently available official reports/statistics, in this case from the UN's Statistics Division to demonstrate both the recent trend in increase of wood fuel use at the household level, plus also the most recently available data (from 2021). This increase in fuelwood use at the household level has had a significant impact on the parameter value fNRB. The density of fuel is not a time bound parameter, hence average of Africa is used as per published data by FAO.

Input parameter	Value	Data source
Wood fuel consumption (m^3/capita/yr)	0.396	FAO[48]
Population of the country	13,276,517	World Bank[49]
Wood fuel consumption (m^3/yr)	**5,257,220**	**Calculated**
Conversion factor (t/m^3)	0.725	FAO[50]
HW (Average consumption of wood fuel tonnes/yr)	**3,811,384**	**Calculated**

Input parameter	Value	Data source
H (in t/year)	6,177,233	Calculated
HW woody biomass consumption in households (in t/year)	3,811,484	Calculated
CE Non domestic woody biomass consumption for energy application (in t/year)	201,877	Calculated, using wood to charcoal conversion factor[51]
NE Non domestic woody biomass consumption for non-energy application (in t/year)	2,163,872	Calculated using biomass expansion factor 1.50
Population of the country	13,276,517	World Bank[52]

Equation (3) of the fNRB tool stipulates that the average household wood fuel consumption is multiplied with the number of households and industries consuming wood fuel for energy and

47 Charcoal transition: https://www.fao.org/3/i6935e/i6935e.pdf
48 https://www.fao.org/faostat/en/#data/FO
49 https://data.worldbank.org/
50 http://www.fao.org/3/a-i6935e.pdf
51 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-33-v2.0.pdf

52 https://data.worldbank.org/

non-energy applications within the country. Since most of the available and reliable data is total wood fuel consumption in households and commercial and other use.

The total population of host country has been sourced as per the published World Bank population data from 2022. This is in line with the fNRB tool.

Step 3: Estimation of quantity of non-renewable biomass (t/year) as per equation (2) of the fNRB tool

NRB = H - RB

Where,

H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

This step calculates the quantity of non-renewable biomass (t/year) (NRB) by subtracting the renewable biomass estimated in step 1 from the total consumption of woody biomass estimated in step 2. No further input parameters are needed under this step.

NRB = 6,177,233– 578,495

NRB = 5,598,738 t/year

Step 4: Estimation of fraction of woody biomass that can be established as non-renewable (fNRB) as per equation (1) of the fNRB tool

$$fNRB = \frac{NRB}{NRB + RB} \qquad \text{Equation (1)}$$

This step calculates the fraction of woody biomass that can be established as non-renewable (fNRB). The non-renewable biomass calculated in step 3 is divided by the sum of non-renewable biomass and renewable biomass calculated in step 1. No further input parameters are needed under this step.

fNRB = 5,598,738/ (5,598,738 + 578,495)

fNRB = **91.0%**

Step 5: Comparison as per para 6 of CDM Tool 30 version 4

As per the Bailis report, the f_{NRB} value for Rwanda is 59%. Rwanda is considered as wood fuel "hotspot" as most of the harvested wood fuel is unsustainable. The resulting f_{NRB} value is higher than the expected f_{NRB} values determined in the Bailis report[53] using the WISDOM method. While direct comparisons between the WISDOM and CDM methodologies are only sometimes appropriate —given the different approaches of the methodologies—the following factors can

53 Bailis R, Drigo R, Ghilardi A and Masera O. 2015. The carbon footprint of traditional woodfuels. Nat Clim Chang 5: 266–272

contribute to variations in the f_{NRB} estimates between the application of the CDM method in the present study and the WISDOM method in the Bailis report[54]:

- **Parameter H (Total consumption of woody biomass in the applicable area in the relevant period (tonnes)):** The current report uses the most recent population statistics (2021) for Rwanda that represents an increase in population numbers since the 2009 data utilised in the Bailis report. Hence, the value of parameter H is higher than the one used Bailis report as it is calculated using the total population.
- Updated 2019 FAO forest products statistics were used in the present study, whereas the comparison study used 2013 FAO forest products statistics.
- **Parameter RB (renewable biomass):** The approach used to determine the amount of forest that is accessible yields lower estimates in the present study than the comparison study. This results in a lower estimated RB and consequently increased NRB and f_{NRB}.
- **Parameter MAI (Mean Annual Increment):** In the present study, MAIs were calculated using a weighted average based on the forest area of three categories (i.e., primary forests, above 20-year secondary forest, below 20- year secondary forest). Data from the 2019 Refinement of the IPCC Guidelines was used in combination with extrapolating the observed forest gain extents between 2000 and 2012 to a future 21-year period. As per the Bailis report, MAI values were derived from a combination of field observations and IPCC values, followed by a different estimation of growth rates as a percentage of standing stock. This approach often yields higher MAIs and may lead to higher estimations of RB and subsequently, lower estimations of NRB and f_{NRB}.

The calculated f_{NRB} value of 0.91 indicates that the consumption of woody biomass from the accessible forest in Rwanda is greater than the country's capacity to renew the supply sustainably. This finding is supported by the consistent and widely-reported deforestation rate and dynamics observed for the country[55,56,57].

The f_{NRB} value of the project was also compared against other registered projects across VERRA and Gold Standard in Rwanda. The fNRB value used by recently registered projects are as follows:

Registry	Project ID	f_{NRB}
Verra	2380	97%
Gold Standard	4261	98%
Gold Standard	2450	98%
Gold Standard	11205	90%
Gold Standard	2449	98%

54 Ibid.

55 FAO. 2020b. Global Forest Resources Assessment 2020: Main Report. Food and Agriculture Organisation of the United Nations, Rome. Available at: https://www.fao.org/forest-resources-assessment/2020/en/

56 Global Forest Watch (GFW). 2023.

57 Mongabay. 2020.

Gold Standard	11098	96.82%
Gold Standard	12308	88.72%

The f_{NRB} value used in other registered projects across various standards range from 88.72% to 98%. As a measure of conservative, the f_{NRB} value for the project has been changed to 88.72% which the lowest among the all the registered projects.

Step 6: Cross check in line with para 13 of CDM Tool 30 version 4.0

In line with para 13 of TOOL 30 V4, the NRB was compared to the top-down product of average above-ground biomass (146.06 t/ha) and the most recent annual deforestation data (3,187 ha/yr). NRB, as calculated in this report according to the CDM Tool 30 v4.0 2022, was found to be 5,598,738 t/yr, which is greater than the cross-check results based on deforestation (465,493 t/yr). It is expected that the cross-check estimate of biomass loss from deforestation could underestimate the extent of non-renewable biomass as it only considers the average biomass stocks from completely deforested areas. Unsustainable wood harvesting resulting in forest degradation and substantial reductions in biomass stocks, but not complete deforestation is not considered in the cross-check. Consequently, the estimate of NRB is considered applicable for this study given that the guidance provided by CDM Tool 30 v4.0 2022 was followed.